The CSP Dichotomy Theorem

A formalization blueprint

Abstract

The Constraint Satisfaction Problem over a finite constraint language is solvable in polynomial time if the language admits a weak near-unanimity polymorphism, and is NP-complete otherwise. This document rewrites Zhuk's simplified proof [10] — together with the prerequisites it imports — as a blueprint for a formalization in Lean 4 and Mathlib. It states precisely what a formalization can and cannot claim (§1); fixes the vocabulary at a granularity a proof assistant can consume (§§2–5); records the theory's interface (§6) and its summit (§7); states the algorithm and the hard half (§§8–9); and reports what Mathlib is missing, what has been built, and what the remaining work is (§§11–12).

Throughout, *formalization notes* mark the places where the informal proof has latitude that a formal one does not: a suppressed side condition, a quantifier whose scope is ambiguous, an object asserted to exist without an argument, or a step whose well-foundedness is left to the reader.

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Introduction

The dichotomy conjecture of Feder and Vardi [4] was settled independently by Bulatov [3] and Zhuk [9] in 2017. Both proofs are long. Zhuk's, in the form published in 2020, is seventy-eight journal pages, and its central induction alternates between global and local information in a way its author later described as forcing "a very complicated induction to prove most of the claims".

In 2024 Zhuk published a simplified proof [10]. The simplification is a redesign of the theory of strong subalgebras so that *every reduction is either strong or global*: for each domain D_x one can build a chain $D_x \supsetneq D_x^{(1)} \supsetneq \cdots \supsetneq \{a\}$ in which every step is either a strong subset or the intersection with a block of an equivalence relation with very strong properties. The types of the intermediate sets need not be tracked — only that such a chain exists, written $C \lhd\!\lhd\!\lhd B$. This is what makes the argument tractable, and it is what makes it a plausible formalization target.

Why this route. Among the available proofs, [10] is the only complete, recent, rigorously written one whose author has already removed the obstruction that made its predecessor unformalizable. Bulatov's proof requires tame congruence theory and centralizer theory, neither of which is formalized anywhere. Restricting to minimal Taylor algebras — which [10] itself suggests would simplify its §2 — turns out to be a bad trade: it would simplify perhaps a quarter of one section and none of the CSP argument, at the cost of importing the cyclic term theorem, which the present route does not otherwise need. Brady's notes [2] are the clearest prose in the subject and are the best source for several of the imported prerequisites, but they stop short of the dichotomy. The route taken here is [10] as the spine, with [2] for the imports, on top of an existing formalization of Zhuk's centre theorem.

What is claimed. Not that the dichotomy has been formalized. §12.5 estimates the transitive closure of what must be proved at 100–130 printed pages, which extrapolates to tens of thousands of lines of Lean — a multi-person-year project, and one that no proof assistant has attempted. What is offered is its design: a route, precise statements, a vocabulary pinned down where the sources leave it loose, an inventory of the missing Mathlib layers, and a Lean development whose foundational modules are complete and whose target theorems are stated.

What the formalization notes are for. A blueprint's value is mostly in its failures of nerve — the places where writing the statement down carefully shows that it does not quite say what it appeared to say. The predecessor project found four such places in twenty-four pages, and all four were in the foundations rather than in the mathematics. The notes here record the same kind of thing: that $\mathbf{Z}_p$ lies in $\mathcal{V}_n$ only when $p \mid n-1$ (Hazard 3.3); that σ^* need not be a congruence (Hazard 3.22); that images do not commute with intersection, and that one proof needs them to (Hazard 3.14); that "we cannot weaken forever" needs a measure that is not the obvious one (Hazard 5.11); that the main induction quantifies over the instance *inside* (Hazard 7.2). §10 collects the harder cases separately: ten places where the source cannot be transcribed at all, of which one — [10]'s restatement of a lemma from [11], with two hypotheses dropped — is false as printed, with a two-line counterexample.

1 What is being proved

1.1 The theorem

Fix a finite set A and a finite set Γ of relations on A, the *constraint language*. An instance of $\mathrm{CSP}(\Gamma)$ is a conjunction $R_1(\bar{x}_1) \wedge \cdots \wedge R_s(\bar{x}_s)$ with each $R_i \in \Gamma$; the question is whether it is satisfiable.

Theorem 1.1 (Dichotomy; Bulatov, Zhuk)**.** *If some weak near-unanimity operation preserves* Γ*, then* $\mathrm{CSP}(\Gamma)$ *is solvable in polynomial time. Otherwise* $\mathrm{CSP}(\Gamma)$ *is* NP*-complete.*

Both halves are conditional on a model of computation, and this is not a formality that a formalization may wave through. “Solvable in polynomial time” quantifies over an encoding of instances as strings, a machine, and a polynomial bound on its step count; “NP-complete” additionally quantifies over all problems in NP and over polynomial-time many-one reductions. Mathlib supplies Turing machines (`Mathlib/Computability/TuringMachine/`) and a notion of computability, but it has *no* complexity class, no polynomial-time predicate, and no notion of polynomial-time reduction. A formalization that claimed Theorem 1.1 verbatim would therefore have to build a complexity layer from scratch, and the mathematics of the dichotomy — which is universal algebra, not complexity theory — would be a minority of the work.

This document separates the two concerns. The mathematical content is stated as propositions that mention no machine; the complexity-theoretic wrapper is isolated in a single, small, clearly marked layer, so that a reader can see exactly which part of Theorem 1.1 rests on the algebra and which on a cost model.

1.2 The layered targets

Write $\mathbf{A} = (A; w)$ for a finite idempotent algebra whose basic operation w is a special weak near-unanimity operation, and $\mathcal{V}_n$ for the class of such algebras with w of arity n (Convention 2.1).

Tractable half. Three layers.

(T0) ***The algebraic core.*** The three theorems of §7 — the existence of a strong subuniverse or a linear congruence, the safety of a reduction to a strong subuniverse, and the codimension-one theorem — together with everything they rest on. These are statements about finite algebras, congruences, and CSP instances regarded as finite combinatorial objects. No computation appears in them.

(T1) ***Correctness of the algorithm.*** Zhuk’s algorithm written as a total function $\textsc{Solve}$ from instances to Booleans, and the proposition

$$\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

This is a statement about a recursive function, and it is provable from (T0) with no cost model. It is the honest formal content of “the algorithm works”.

(T2) ***The complexity wrapper.*** That $\textsc{Solve}$ runs in polynomial time in the size of the instance, for fixed Γ. This requires a cost model and is the only place where one is needed.

Hard half. Two layers.

(H0) ***The algebraic core.*** If no weak near-unanimity operation preserves Γ then Γ primitively positively interprets every finite constraint language — in particular one whose

CSP is NP-hard. This is again pure algebra: the Galois connection between polymorphisms and primitive positive definability, the Maróti–McKenzie theorem that a Taylor term yields a weak near-unanimity term, and the Bulatov–Jeavons–Krokhin reduction.

(H1) ***The complexity wrapper.*** That a primitive positive interpretation yields a polynomial-time many-one reduction, and hence NP-hardness. Again a cost model, and again small.

The claim this document makes is that (T0), (T1) and (H0) are the mathematics, that they are what the literature actually proves, and that (T2) and (H1) are a separable engineering layer whose difficulty is unrelated to the dichotomy. The blueprint is written for (T0), (T1) and (H0); §11 states (T2) and (H1) precisely and says what building them would cost, without proving them.

Remark 1.2. The separation is not a dodge. Layer (T1) is strictly stronger than " CSP(Γ) is decidable", which is trivial for a finite template: it says that one particular, highly non-obvious algorithm — the one whose existence is the content of the dichotomy — is correct. Everything that makes the dichotomy hard lives in (T0). Conversely, no part of (T0)–(T1) becomes easier if a cost model is available.

2 Standing conventions

Zhuk's papers carry their hypotheses in prose: "in this paper we assume that every algebra is a finite idempotent algebra having a WNU term operation". A formalization cannot carry a hypothesis in prose, and an informal reader cannot audit which results actually consume which hypothesis. We therefore fix the hypotheses as a labelled convention and record, for each one, where it is consumed.

Convention 2.1 (Standing hypotheses)**.** Throughout, $n \geq 3$ is a fixed integer and:

(a) every algebra is *finite* and has a *nonempty* universe;
(b) every algebra has exactly one basic operation, of arity n, written w;
(c) w is *idempotent*: $w(x, \ldots, x) = x$;
(d) w is a *weak near-unanimity* operation: $w(y, x, \ldots, x) = w(x, y, x, \ldots, x) = \cdots = w(x, \ldots, x, y)$ for all x, y;
(e) w is *special*: $w(x, \ldots, x, y) = w(x, \ldots, x, w(x, \ldots, x, y))$ for all x, y.

The class of algebras satisfying a–e is written $\mathcal{V}_n$. Since only finite algebras are considered, $\mathcal{V}_n$ is not a variety.

Remark 2.2 (Why one basic operation)**.** Restricting to a single basic operation is not a loss. The dichotomy hypothesis is that *some* weak near-unanimity operation preserves Γ; one may then discard the original signature and work in the algebra $(A; w)$ whose only basic operation is that one, because every relation of Γ is a subuniverse of a power of $(A; w)$ and that is all the argument uses. Passing from a WNU w to a *special* one is Lemma 6.20, and costs an increase of arity from n to $n^{n!}$.

The gain is large for a formalization: a term is a finite tree with n-ary branching over a single symbol, so "term operation" needs no signature bookkeeping, and $\mathrm{Clo}(\mathbf{A})$ is the clone generated by one operation.

Formalization note 2.3 (Where each hypothesis is consumed)**.** a is used in every induction on $|A|$ and in every minimality argument (a minimal element of a nonempty family of subsets exists); it cannot be dropped anywhere. c is what makes singletons subuniverses and makes Sg of a nonempty set nonempty. d is consumed only through the results it implies — the existence of absorption-theoretic structure — and never directly in §§5–7. e is used where a unary polynomial must be idempotent under iteration, chiefly in the theory of Sg-closures of two-element sets. A formalization should carry a–e as typeclass instances on a structure and not as ambient hypotheses, so that the audit is mechanical.

Convention 2.4 (The empty set)**.** A *subuniverse* of $\mathbf{A}$ is a subset closed under w; the empty set is a subuniverse under this definition, and we allow it. A *subalgebra* is a nonempty subuniverse; $\mathbf{B} \leq \mathbf{A}$ always carries $B \neq \varnothing$.

The relations $<_T$, $\leq_T$ and $\lhd\lhd\lhd$ of §4 are defined only between *nonempty* sets; in particular $\varnothing \lhd\lhd\lhd A$ never holds. Where a construction can produce the empty set — a projection of an intersection, most often — the dotted variants $\dot{<}_T$, $\dot{\leq}_T$, $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ are used, and they mean "the stated relation holds, or the left-hand side is empty".

Formalization note 2.5 (The dot is not decoration)**.** Conflating $\lhd\lhd\lhd$ with $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ is the single most common way to misread §4. Every propagation statement of §6 that projects or intersects produces a dotted conclusion, and every statement that consumes a $\lhd\lhd\lhd$ hypothesis needs the undotted one. In a formalization the two should be distinct constants, with the dotted one *defined* as `C = ∅ ∨ C ◁◁◁ B`, so that the case split is forced at every use site rather than being resolved by the reader's judgement.

Convention 2.6 (Relations, and indices)**.** A relation is a subset of a product $A_1 \times \cdots \times A_m$ of universes, indexed by $[m] = \{1, \ldots, m\}$; $\mathbf{R} \le \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ means additionally that R is a subuniverse of the product algebra. We write $\mathrm{pr}_{i_1,\ldots,i_s}(R)$ for the projection onto the listed coordinates and $\mathbf{R} \le_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ when every $\mathrm{pr}_i(R) = A_i$.

Products are indexed by *arbitrary finite index sets*, not only by initial segments of $\mathbb{N}$. Several constructions in §7 form a product over the variable set of an instance, or over a set of subuniverses, and reindexing those to $[m]$ by hand is exactly the kind of bookkeeping that a formalization pays for repeatedly and that buys nothing.

3 Vocabulary

This section fixes the universal-algebraic vocabulary. It is Zhuk's §2.1, rewritten so that every definition is unambiguous when read by a machine. The substantive content begins in §4; a reader who knows the subject should skim to Definition 3.21 (irreducible congruences) and Definition 3.23 (bridges), which are the two places where the conventions genuinely matter.

3.1 Algebras, subuniverses, terms

Definition 3.1 (Algebra)**.** An algebra is a pair $\mathbf{A} = (A; w^{\mathbf{A}})$ with A a finite nonempty set and $w^{\mathbf{A}} \colon A^n \to A$, subject to Convention 2.1. We write A for the universe of $\mathbf{A}$ and drop the superscript on w when no confusion arises.

Definition 3.2 (The algebras $\mathbf{Z}_p$)**.** For a prime p with $p \mid n-1$, $\mathbf{Z}_p$ is the algebra on $\{0, 1, \ldots, p-1\}$ with $w^{\mathbf{Z}_p}(x_1, \ldots, x_n) = x_1 + \cdots + x_n \bmod p$.

Formalization note 3.3 ($\mathbf{Z}_p$ is only sometimes in $\mathcal{V}_n$)**.** Zhuk defines $\mathbf{Z}_p$ for every prime p and then asserts that "every algebra $\mathbf{Z}_p$ belongs to $\mathcal{V}_n$". That assertion carries a silent side condition. The operation $x_1 + \cdots + x_n \bmod p$ is idempotent exactly when $n \equiv 1 \pmod p$: idempotence says $nx \equiv x$ for all x, i.e. $p \mid n-1$. Given that, it is also a WNU (all the identities read $(n-1)x + y = y$) and special (both sides reduce to y).

So $\mathbf{Z}_p \in \mathcal{V}_n$ if and only if $p \mid n-1$, and every statement of the form "$\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for some prime p" (Lemma 4.14, Definition 3.28) silently produces a p dividing $n-1$. A formalization must decide whether $\mathbf{Z}_p$ is defined for all p and merely fails to lie in $\mathcal{V}_n$ for bad p, or is defined only for $p \mid n-1$. The first is cleaner; the second makes the membership free. Either way the side condition must be visible, because "for some prime p" otherwise reads as an unconstrained existential.

Definition 3.4 (Subuniverse, subalgebra)**.** $B \subseteq A$ is a *subuniverse* of $\mathbf{A}$ if $w(b_1, \ldots, b_n) \in B$ for all $b_1, \ldots, b_n \in B$. If moreover $B \neq \varnothing$ then $\mathbf{B} = (B; w\restriction_{B^n})$ is a *subalgebra*, written $\mathbf{B} \le \mathbf{A}$.

Definition 3.5 (Term operations)**.** $\mathrm{Clo}(\mathbf{A})$ is the least set of finitary operations on A containing all projections and closed under composition with w. Its m-ary part is written $\mathrm{Clo}_m(\mathbf{A})$. Elements of $\mathrm{Clo}(\mathbf{A})$ are the *term operations* of $\mathbf{A}$.

Formalization note 3.6 (Terms versus term operations)**.** Zhuk writes $t \in \mathrm{Clo}(\mathbf{A})$ and treats t as both a syntactic term and the operation it induces. These must be distinguished in a formalization: a term is a finite tree, and distinct terms may induce the same operation. Every statement below that quantifies over $\mathrm{Clo}(\mathbf{A})$ is a statement about *operations*, so the syntactic layer may be quotiented away — but it cannot be skipped, because Sg is characterised through terms (Lemma 3.8) and because the arity bookkeeping in the absorption arguments is syntactic.

Definition 3.7 (Generation)**.** For $X \subseteq A^m$, $\mathrm{Sg}_{\mathbf{A}}(X)$ is the least subuniverse of $\mathbf{A}^m$ containing X.

Lemma 3.8 (Generation by terms)**.** *Let $X = \{x_1, \ldots, x_k\} \subseteq A^m$ be nonempty. Then*

$$\mathrm{Sg}_{\mathbf{A}}(X) = \{\, t^{\mathbf{A}^m}(x_1, \ldots, x_k) \;:\; t \in \mathrm{Clo}_k(\mathbf{A}) \,\}.$$

The generator list is fixed*: the same list $x_1, \ldots, x_k$ in the same order serves every element of the closure, with t varying.*

Formalization note 3.9. The "fixed list" phrasing is what makes Lemma 3.8 usable. The weaker reading — for each y in the closure there are *some* generators and *some* term — is also true but useless, because every later application applies one term simultaneously to several closure elements and needs their generator lists to coincide. This was a defect found in review of the predecessor blueprint and is called out here for the same reason.

3.2 Relations

Definition 3.10 (Relational vocabulary)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) R is *subdirect* if $\mathrm{pr}_i(R) = A_i$ for every i; $\mathbf{R} \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ abbreviates "R is a subdirect subuniverse".
(b) For $R \subseteq A^m$: R is *reflexive* if $(a, \ldots, a) \in R$ for every $a \in A$.
(c) For binary $\delta_1 \subseteq A_1 \times A_2$, $\delta_2 \subseteq A_2 \times A_3$: $\delta_1 \circ \delta_2 = \{(a, c) : \exists b\ (a, b) \in \delta_1 \wedge (b, c) \in \delta_2\}$, and $\delta^{-1} = \{(b, a) : (a, b) \in \delta\}$.
(d) For $B \subseteq A_1$ and $\delta \subseteq A_1 \times A_2$: $B \circ \delta = \{c : \exists b \in B\ (b, c) \in \delta\}$.
(e) A subdirect $\delta \subseteq A \times B$ is *linked* if the bipartite graph on $A \sqcup B$ with edge set δ is connected. It is *bijective* if $|\delta| = |A| = |B|$.
(f) For binary σ and $m \geq 2$, $\sigma^{[m]} = \{(a_1, \ldots, a_m) : (a_i, a_j) \in \sigma\ \forall i, j\}$.

Formalization note 3.11 (Composition is diagrammatic)**.** $\delta_1 \circ \delta_2$ applies δ_1 *first*. This is the order of `Rel.comp` and the opposite of `Function.comp`. Every composite of bridges and of congruences below is written left to right, and silently reversing it produces statements that are true-looking and wrong — in Corollary 6.14(l) the composite $\sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$ is asymmetric in k and ℓ and the order is the content.

Formalization note 3.12 ("Linked" is used for two different things)**.** Zhuk uses *linked* both for a binary relation (Definition 3.10(e)) and for a CSP instance (Definition 5.8). They are related but not the same predicate, and the coincidence of names causes real confusion in §7, where both occur in one proof. Use two names.

Definition 3.13 (Quotients of relations)**.** For an equivalence σ on A and $a \in A$, a/σ is the class of a; for $B \subseteq A$, $B/\sigma = \{b/\sigma : b \in B\}$; for $R \subseteq A^m$, $R/\sigma = \{(b_1/\sigma, \ldots, b_m/\sigma) : \bar{b} \in R\}$.

Formalization note 3.14 (Images do not commute with intersection)**.** B/σ and R/σ are *images*. Three consequences that the source navigates silently:

(a) B/σ is a subset of A/σ, not the quotient of B by $\sigma \cap B^2$. For B a subuniverse and σ a congruence the two are canonically isomorphic as algebras, and the source moves between them freely. Fix B/σ to be the image and prove the isomorphism once.
(b) In general $(C_1 \cap C_2)/\sigma \subsetneq C_1/\sigma \cap C_2/\sigma$. Lemma 6.4(fm) needs equality for a family $C_1, \ldots, C_t$, and gets it from the extra hypothesis $(C_i \circ \sigma) \cap B = C_i$, i.e. that each C_i is σ-saturated inside B. That is a content step, not notation.
(c) R/σ need not be stable under anything, and need not be a subuniverse of $(\mathbf{A}/\sigma)^m$ unless σ is a congruence.

3.3 Rectangularity

Definition 3.15 (Parallelogram property, rectangularity)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) The i-th variable of R is *rectangular* if for all $\bar{a}, \bar{b}$,

$$\bar{a}[i \mapsto b_i] \in R, \quad \bar{b}[i \mapsto a_i] \in R, \quad \bar{b} \in R \implies \bar{a} \in R,$$

where $\bar{a}[i \mapsto c]$ is $\bar{a}$ with its i-th entry replaced by c. R is *rectangular* if every variable is.

(b) *R has the parallelogram property* if for every permutation of the coordinates giving R' and every $\ell \in [m-1]$,

$$(a_1, \dots, a_\ell, b_{\ell+1}, \dots, b_m) \in R', \quad (b_1, \dots, b_\ell, a_{\ell+1}, \dots, a_m) \in R', \quad \bar{b} \in R' \implies \bar{a} \in R'.$$

(c) The *rectangular closure* of R is the least rectangular relation containing R.

Lemma 3.16 (Two assertions the source makes in passing)**.** (a) *A relation with the parallelogram property is rectangular.*

(b) *The rectangular closure exists.*

Proof. (a) Apply the parallelogram property to the permutation carrying coordinate i to position 1, with $\ell = 1$. (b) Rectangularity at coordinate i is a Horn implication, so the rectangular relations containing R are closed under arbitrary intersection and the family is nonempty (it contains $A_1 \times \cdots \times A_m$); the intersection of all of them is the least one. □

Formalization note 3.17. Both parts are asserted without proof in [10] — (a) as "as it follows from the definitions", (b) by the definite article in "*the* minimal rectangular relation". Both are easy and both must be present. Note that "R is rectangular" unqualified — the form used in "rectangular closure" — is never defined in the source; it means every variable is rectangular.

Formalization note 3.18 (Quantifying over permutations)**.** The parallelogram property is stated "any permutation of its variables gives a relation R' satisfying …". Formally this is a quantification over $\mathfrak{S}_m$ combined with a quantification over the split point ℓ; equivalently, and more usably, over all ways of splitting $[m]$ into two nonempty blocks. The second formulation is the one to formalize: it avoids transporting R along a permutation, which in a dependently typed setting means an equivalence of index types and a coercion at every use.

3.4 Congruences

Definition 3.19 (Stability)**.** Let $R \subseteq A_1 \times \cdots \times A_m$ and let σ be a binary relation on A_i. The i-th variable of R is *stable under* σ if $\bar{a} \in R$ and $(a_i, b) \in \sigma$ imply $\bar{a}[i \mapsto b] \in R$. R is *stable under* σ if every variable is (each with respect to σ on the corresponding factor, when the factors agree).

Definition 3.20 (Congruence)**.** A *congruence* on $\mathbf{A}$ is an equivalence relation on A that is a subuniverse of $\mathbf{A} \times \mathbf{A}$. It is *nontrivial* if it is neither the equality relation $0_{\mathbf{A}}$ nor A^2. The quotient $\mathbf{A}/\sigma$ is the algebra on A/σ with $w(\bar{a}/\sigma) = w(\bar{a})/\sigma$.

Definition 3.21 (Irreducible congruence and its cover)**.** A congruence σ on $\mathbf{A}$ is *irreducible* if it cannot be written as an intersection of binary subuniverses of $\mathbf{A} \times \mathbf{A}$ each stable under σ and each properly containing σ. Equivalently: there are no $\mathbf{S}_1, \dots, \mathbf{S}_k \leq \mathbf{A}/\sigma \times \mathbf{A}/\sigma$ with $0_{\mathbf{A}/\sigma} = S_1 \cap \cdots \cap S_k$ and $S_i \neq 0_{\mathbf{A}/\sigma}$ for every i.

For irreducible σ, σ^* denotes the least $\delta \leq \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ.

Formalization note 3.22 (Three traps in Definition 3.21)**.** (a) Irreducibility is *not* meet-irreducibility in the congruence lattice. The $\mathbf{S}_i$ range over binary *subuniverses* stable under σ, which need not be equivalence relations. The two notions differ, and every use below is the stated one.

(b) σ^* need not be a congruence. It is a subuniverse of $\mathbf{A}^2$ stable under σ, nothing more; that σ^* *is* a congruence is an extra hypothesis in the definition of a linear congruence (Definition 4.8), which would be redundant otherwise.

(c) Existence of σ^* needs an argument: the family of $\delta \supsetneq \sigma$ stable under σ is nonempty (it contains A^2) and, by irreducibility, closed under the intersections that matter, so a least element exists. Uniqueness is exactly what irreducibility buys; without it there is no "the" cover. A formalization must produce σ^* as data attached to a proof of irreducibility, not as a standalone function.

3.5 Bridges

Definition 3.23 (Bridge)**.** Let σ_1, σ_2 be congruences on $\mathbf{D}_1, \mathbf{D}_2$. A relation $\delta \leq \mathbf{D}_1^2 \times \mathbf{D}_2^2$ is a *bridge from σ_1 to σ_2* if

(a) the first two variables of δ are stable under σ_1;
(b) the last two variables of δ are stable under σ_2;
(c) $\mathrm{pr}_{1,2}(\delta) \supsetneq \sigma_1$ and $\mathrm{pr}_{3,4}(\delta) \supsetneq \sigma_2$;
(d) $(a_1, a_2, a_3, a_4) \in \delta$ implies $\big((a_1, a_2) \in \sigma_1 \iff (a_3, a_4) \in \sigma_2\big)$.

For a bridge δ put $\widetilde{\delta} = \{(x, y) : (x, x, y, y) \in \delta\}$. A bridge $\delta \subseteq D^4$ is *reflexive* if $(a, a, a, a) \in \delta$ for all $a \in D$.

Remark 3.24 (The motivating example)**.** On $\mathbb{Z}_4$, the relation $\delta = \{(a_1, a_2, a_3, a_4) : a_1 - a_2 = 2a_3 - 2a_4\}$ is a bridge from $0_{\mathbb{Z}_4}$ to congruence mod 2. Condition (d) is the content: $a_1 = a_2$ exactly when $a_3 \equiv a_4 \pmod 2$. Bridges are the formal shadow of a centralizer relation; see [8].

Definition 3.25 (Composition of bridges)**.** For bridges δ_1 from σ_0 to σ_1 and δ_2 from σ_1 to σ_2, set

$$(\delta_1 * \delta_2)(x_1, x_2, z_1, z_2) = \exists y_1 \exists y_2\ \delta_1(x_1, x_2, y_1, y_2) \wedge \delta_2(y_1, y_2, z_1, z_2).$$

Lemma 3.26 (Bridge composition; [9, Lem. 6.3])**.** *If $\sigma_0, \sigma_1, \sigma_2$ are irreducible then $\delta_1 * \delta_2$ is a bridge from σ_0 to σ_2, and $\widetilde{\delta_1 * \delta_2} = \widetilde{\delta_1} \circ \widetilde{\delta_2}$.*

Formalization note 3.27. Lemma 3.26 is imported, not proved, in [10]; it is [9, Lemma 6.3]. Both conclusions are used: the first in every chain of bridges, the second — the identity $\widetilde{\cdot}$ commutes with composition — in Lemma 6.22, where a bridge with $\widetilde{\delta}$ linked is powered up until $\widetilde{\delta} = A^2$. The irreducibility hypothesis on *all three* congruences is needed and is easy to drop by accident.

Definition 3.28 (Perfect linear congruence)**.** A congruence σ on $\mathbf{A}$ is a *perfect linear congruence* if it is irreducible and there are a prime p and $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ with $\mathrm{pr}_{1,2}(\zeta) = \sigma^*$ and

$$(a_1, a_2, b) \in \zeta \implies \big((a_1, a_2) \in \sigma \iff b = 0\big).$$

Perfect linear congruences are the payoff of the whole bridge machinery: they let the passage from σ to σ^* be tracked by a single element of $\mathbf{Z}_p$, which is what turns a CSP instance into a system of linear equations in §7.3.

4 The six types of strong subuniverse

This section is Zhuk's §2.2. It defines six binary relations $C <_T^A B$ between subsets of a fixed algebra $\mathbf{A}$, one for each type T, and the transitive-closure relation $C \lhd\lhd\lhd^A B$. Everything in §§6–7 is expressed in this language, so the definitions repay care.

The informal reading is: $C <_T^A B$ says that C is a *strong* subset of B — strong enough that the CSP algorithm may reduce a variable's domain from B to C without losing all solutions — and T records *why* it is strong. Three of the reasons are absorption-theoretic (BA, C, S) and three are congruence-theoretic (D, L, PC).

4.1 Absorption

Definition 4.1 (Absorbing subuniverse)**.** A subuniverse B of $\mathbf{A}$ is *absorbing* if there is $t \in \mathrm{Clo}_k(\mathbf{A})$, for some k, such that for every $i \in [k]$

$$t(B, \dots, B, \underbrace{A}_{i}, B, \dots, B) \subseteq B.$$

We then say B *absorbs* $\mathbf{A}$ *with the term* t. If t may be chosen binary, B is a *binary absorbing* subuniverse (BA); if ternary, a *ternary absorbing* subuniverse.

Formalization note 4.2 (The tuple form)**.** Definition 4.1 must be formalized over *tuples constrained coordinatewise*, not as "for all $b_1, \dots, b_k \in B$ and $a \in A$, $t(b_1, \dots, b_{i-1}, a, b_{i+1}, \dots, b_k) \in B$". The two agree, but the second phrasing invites the variant in which b_i is quantified and then discarded, which is vacuously true when $B = \varnothing$ and at $k = 1$ where the intended content is not vacuous. This was an actual defect in the predecessor blueprint. State it as: for every $\bar{a} \in A^k$ such that $a_j \in B$ for all $j \neq i$, one has $t(\bar{a}) \in B$.

Definition 4.3 (Central subuniverse)**.** A subuniverse C of $\mathbf{A}$ is *central* (C) if it is absorbing and for every $a \in A \setminus C$,

$$(a, a) \notin \mathrm{Sg}_{\mathbf{A}}((\{a\} \times C) \cup (C \times \{a\})).$$

Lemma 4.4 ([11, Cor. 6.11.1])**.** *A central subuniverse is a ternary absorbing subuniverse.*

Remark 4.5 (Already formalized)**.** Lemma 4.4 is the one deep import of this theory that is already available in Lean: it is the main theorem of the predecessor project (`zhuk_main` in `ZhukLean`), proved there from the definitions via the Barto–Kazda relational description of absorption and the Zhuk–Kozik doubling trick. See §12.1.

Definition 4.6 (BA and centre free; S)**.** $\mathbf{A}$ is *BA and centre free* if it has no proper nonempty binary absorbing subuniverse and no proper nonempty central subuniverse.

For $\varnothing \neq C \lneq B \leq A$ write $C <_{\mathrm{S}}^A B$ if there is a subuniverse $D \subseteq C$ of $\mathbf{B}$ that is simultaneously binary absorbing and central in $\mathbf{B}$. $\mathbf{A}$ is S*-free* if there is no $C <_{\mathrm{S}} \mathbf{A}$; equivalently, no nonempty proper $D \leq A$ is both binary absorbing and central.

Formalization note 4.7 (S is not the conjunction of BA and C)**.** $C <_{\mathrm{S}} B$ does not say that C is binary absorbing and central in $\mathbf{B}$; it says that C *contains* some D that is. The type S exists precisely so that it is closed under enlarging C, which BA and C are not. Note also the asymmetry with S-freeness, which *is* stated as a conjunction. Getting this backwards makes Lemma 6.4(ft) unprovable.

4.2 Linear and PC congruences

Definition 4.8 (Linear congruence)**.** A congruence σ on $\mathbf{A}$ is *linear* if

(a) σ is irreducible;
(b) σ^* is a congruence;
(c) there are a prime p and $S \leq (\sigma^*)^{[4]}$ such that for every block B of σ^* there is $m \geq 0$ with

$$(B/\sigma;\ S \cap (B/\sigma)^4) \ \cong\ (\mathbb{Z}_p^m;\ x_1 - x_2 = x_3 - x_4).$$

An irreducible congruence that is not linear is a *PC congruence.*

The relation S of (c) is a bridge from σ to σ with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$. This is the useful reformulation:

Lemma 4.9 ([10, Lem. 4.9]). *For an irreducible congruence σ the following are equivalent:*

(a) *σ is linear;*
(b) *there is a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$.*

Formalization note 4.10 (Definition (c) versus criterion (b)). Definition 4.8(c) is a per-block isomorphism statement with an existential m *depending on the block*, and it quantifies over blocks of σ^*, not of σ. Lemma 4.9 replaces all of that by one bridge. A formalization should take (b) as the *definition* and prove (c) as a structure theorem, reversing the paper's order: (b) is a Σ_1 statement over finite data and is what every consumer actually uses, while (c) drags in a quotient, a choice of prime, and a family of isomorphisms.

The dichotomy "linear or PC" is then *by definition* exhaustive on irreducible congruences, which is how the paper uses it. Note that PC-ness is a negative condition; every lemma about PC congruences is proved by contradiction against Lemma 4.9(b).

Lemma 4.11 (No bridge crosses the types). *If σ_1 is linear, σ_2 is irreducible, and there is a bridge from σ_1 to σ_2, then σ_2 is linear.*

Lemma 4.12 (Bridges from PC congruences are trivial). *Let σ be a PC congruence on A and δ a reflexive bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta) = \sigma^*$. Then*

$$\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4) \quad \text{or} \quad \delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_4) \wedge \sigma(x_2, x_3).$$

Lemma 4.13 (Bridges into PC congruences). *Let δ be a bridge from a PC congruence σ_1 on $\mathbf{A}_1$ to an irreducible congruence σ_2 on $\mathbf{A}_2$, with $\mathrm{pr}_{1,2}(\delta) = \sigma_1^*$ and $\mathrm{pr}_{3,4}(\delta) = \sigma_2^*$. Then*

(a) *σ_2 is a PC congruence;*
(b) *$\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$;*
(c) *$\{(a/\sigma_1, b/\sigma_2) : (a, b) \in \widetilde{\delta}\}$ is bijective;*
(d) *$\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_3) \wedge \widetilde{\delta}(x_2, x_4)$ or $\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_4) \wedge \widetilde{\delta}(x_2, x_3)$.*

Two results connect the new vocabulary to the linear and PC subuniverses of the original proof.

Lemma 4.14. *If σ is a linear congruence on $\mathbf{A} \in \mathcal{V}_n$ with $\sigma^* = A^2$, then $\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for a prime p.*

Lemma 4.15. *If σ is a PC congruence on $\mathbf{A}$ with $\sigma^* = A^2$, then $\mathbf{A}/\sigma$ is polynomially complete.*

Formalization note 4.16 (Polynomial completeness). An algebra is *polynomially complete* if the clone generated by its basic operations together with all constants is the clone of *all* operations on its universe. Lemma 4.15 is where the name "PC" comes from, and it is the only place in the development where polynomial completeness is used as more than a label. Formalizing it needs the Istinger–Kaiser characterization [5]; formalizing everything else needs only the *name*. A staged development should therefore treat "PC congruence" as "irreducible and not linear" throughout, and prove Lemma 4.15 last or not at all — it is used only to connect with [9], not by any argument here.

4.3 The types

Definition 4.17 (The six types)**.** Let $\varnothing \neq C \lneq B \leq A$. We write:

- $C <^A_{\text{BA}} B$ if C is a binary absorbing subuniverse of $\mathbf{B}$;
- $C <^A_{\text{C}} B$ if C is a central subuniverse of $\mathbf{B}$;
- $C <^A_{\text{D}} B$ if there is an irreducible congruence σ on $\mathbf{A}$ with
  - (i) $B^2 \subseteq \sigma^*$,
  - (ii) $C = B \cap E$ for some block E of σ,
  - (iii) $\mathbf{B}/\sigma$ is BA and centre free;
- $C <^A_{\text{L}} B$ if $C <^A_{\text{D}} B$ with the witnessing σ linear;
- $C <^A_{\text{PC}} B$ if $C <^A_{\text{D}} B$ with the witnessing σ a PC congruence;
- $C <^A_{\text{S}} B$ as in Definition 4.6.

When the witnessing congruence matters we write $C <^A_{T(\sigma)} B$; for $T \in \{\text{BA}, \text{C}, \text{S}\}$, where no congruence is involved, σ is taken to be A^2.

Formalization note 4.18 (The superscript A is not decoration)**.** For $T \in \{\text{D}, \text{L}, \text{PC}\}$ the witnessing congruence lives on $\mathbf{A}$, not on $\mathbf{B}$, and condition (i) relates the two: $B^2 \subseteq \sigma^*$ says B sits inside a single block of σ^*. This is why the relation carries the ambient algebra as a superscript. For $T \in \{\text{BA}, \text{C}, \text{S}\}$ the superscript is genuinely irrelevant and Zhuk drops it. A formalization should not overload: define $<^A_T$ with A always present, and prove the independence for the three absorption types as a lemma.

Note also that $\mathbf{B}/\sigma$ in (iii) means B/σ with the induced algebra structure, which requires B to be a subuniverse and σ restricted to B; since $B^2 \subseteq \sigma^*$ and not necessarily $B^2 \subseteq \sigma$, the quotient B/σ is in general *not* a single point, and its being BA and centre free is a real condition.

Definition 4.19 (Multi-types)**.** For $T \in \{\text{L}, \text{PC}, \text{D}\}$ write $C <^A_{\mathcal{M}T} B$ if $C \neq \varnothing$ and $C = C_1 \cap \cdots \cap C_t$ with $C_i <^A_T B$ for every $i \in [t]$.

Definition 4.20 (The strong-reduction relation)**.** Write $C \lhd\!\lhd\!\lhd^A B$ if there are $B_0, \ldots, B_m \subseteq B$ and $T_1, \ldots, T_m \in \{\text{BA}, \text{C}, \text{S}, \text{D}\}$ with

$$C = B_m <^A_{T_m} B_{m-1} <^A_{T_{m-1}} \cdots <^A_{T_1} B_0 = B.$$

Here $m = 0$ is allowed, so $\lhd\!\lhd\!\lhd^A$ is reflexive. A congruence *comes from* $C \lhd\!\lhd\!\lhd^A B$ if it is one of the witnessing congruences in such a chain. We write $B \lhd\!\lhd\!\lhd A$ for $B \lhd\!\lhd\!\lhd^A A$, and $C \leq^A_{T(\sigma)} B$ for "$C = B$ or $C <^A_{T(\sigma)} B$".

Formalization note 4.21 ($\lhd\!\lhd\!\lhd$ is existential over chains)**.** $C \lhd\!\lhd\!\lhd^A B$ asserts the *existence* of a chain; the intermediate types are forgotten. This is deliberate — it is the paper's central simplification over [9], where the types in the middle had to be tracked — and it is why $\lhd\!\lhd\!\lhd$ is usable as a hypothesis. In Lean it is the reflexive transitive closure of $\bigcup_{T \in \{\text{BA},\text{C},\text{S},\text{D}\}} <^A_T$, i.e. `Relation.ReflTransGen`, and the induction principle that comes with that is exactly the one the proofs use. Note that L and PC are *absent* from the list because D subsumes them.

Convention 4.22 (Dotted variants)**.** $C \dot{<}^A_T B$, $C \dot{\leq}^A_T B$ and $C \;\dot{\lhd}\!\dot{\lhd}\!\dot{\lhd}^A\; B$ mean that the undotted relation holds *or* $C = \varnothing$. See Hazard 2.5.

5 Instances

This section is Zhuk's §3.1. It is entirely combinatorial: no algebra is used beyond the fact that each domain carries a w and each constraint relation is a subuniverse of a product of domains. It is also where a formalization must make the most decisions, because the paper's instances are informal syntactic objects manipulated by operations — weakening, covering, renaming — that are described in words.

5.1 Instances and solutions

Definition 5.1 (Instance)**.** Fix a finite set V of variables and, for each $x \in V$, an algebra $\mathbf{D}_x \in \mathcal{V}_n$. A *constraint* is a pair $C = (\bar{x}, R)$ where $\bar{x} = (x_1, \dots, x_m)$ is a tuple of variables and $R \leq \mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$. An *instance* $\mathcal{I}$ is a finite list of constraints; we write $C \in \mathcal{I}$, and $\mathrm{Var}(C)$, $\mathrm{Var}(\mathcal{I})$ for the variables occurring. A *subinstance* is a sublist.

A *solution* is a map s with $s(x) \in D_x$ for all x such that $(s(x_1), \dots, s(x_m)) \in R$ for every constraint. The solution set of $\mathcal{I}$ is *subdirect* if for every x and every $a \in D_x$ there is a solution with $s(x) = a$.

Formalization note 5.2 (Lists, not sets; tuples, not sets of variables)**.** Two representation choices are forced by the way the paper argues, and both differ from the naive one.

Constraints form a list, not a set. Proofs delete "a constraint $C \in \mathcal{I}$" and reason about $\mathcal{I} \setminus \{C\}$, and the same relation may occur twice on different variable tuples — and, in the expanded coverings of §5.5, the same relation on the *same* tuple can legitimately occur twice. Multiset or list semantics is required; `List` with membership by index is the safe choice.

A constraint's scope is a tuple, not a set. Repeated variables in a scope are allowed and occur — $\zeta(x, x', z)$ in §7 arises by splitting one variable into two, and $R(x, x, \dots, x)$ appears explicitly in Definition 5.14. So $\mathrm{Var}(C)$ is the image of the scope, and $|\mathrm{Var}(C)|$ may be smaller than the arity.

Definition 5.3 (Relations defined by an instance)**.** For $x_1, \dots, x_m \in \mathrm{Var}(\mathcal{I})$, $\mathcal{I}(x_1, \dots, x_m)$ is the set of $(a_1, \dots, a_m)$ such that $\mathcal{I}$ has a solution with $s(x_i) = a_i$ for all i. It is a subuniverse of $\mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$, being defined by a primitive positive formula over the constraint relations.

5.2 Reductions

Definition 5.4 (Reduction)**.** A *reduction* for $\mathcal{I}$ is a map $D^{(\top)}$ assigning to each $x \in \mathrm{Var}(\mathcal{I})$ a subuniverse $D_x^{(\top)} \subseteq D_x$, possibly empty. It is *nonempty* if every $D_x^{(\top)} \neq \varnothing$. The instance $\mathcal{I}^{(\top)}$ is $\mathcal{I}$ with each domain replaced by $D_x^{(\top)}$ and each constraint relation intersected with the corresponding box. For reductions we write $D^{(\perp)} \lhd\lhd\lhd\, D^{(\top)}$ and $D^{(\perp)} \leq_T D^{(\top)}$ when the corresponding relation holds at every variable.

Formalization note 5.5 (Empty reductions)**.** Reductions are allowed to be empty at some or all variables — the maximal 1-consistent reduction below $D^{(B,\top)}$ in the proof of Theorem 7.1 is constructed exactly so that it may come out empty, and the case split on whether it does is the branch point of the whole argument. But $D^{(\perp)} \lhd\lhd\lhd\, D^{(\top)}$ requires nonemptiness at every variable (Convention 2.4). Both notions are needed and they must be different objects.

5.3 Consistency

Definition 5.6 (Paths)**.** A *path* in $\mathcal{I}$ is an alternating sequence $z_1 - C_1 - z_2 - \cdots - C_{l-1} - z_l$ with $z_i, z_{i+1} \in \mathrm{Var}(C_i)$. It *connects b and c* if there are $a_i \in D_{z_i}$ with $a_1 = b$, $a_l = c$, and $(a_i, a_{i+1}) \in \mathrm{pr}_{z_i, z_{i+1}}(C_i)$ for every i. $\mathcal{I}$ is a *tree-instance* if it has no path with $l \geq 3$, $z_1 = z_l$ and $C_1, \dots, C_{l-1}$ pairwise distinct.

Definition 5.7 (Consistency). $\mathcal{I}$ is 1-*consistent* if $\mathrm{pr}_z(C) = D_z$ for every constraint C and every $z \in \mathrm{Var}(C)$. A reduction $D^{(\top)}$ is 1-consistent for $\mathcal{I}$ if $\mathcal{I}^{(\top)}$ is 1-consistent. $\mathcal{I}$ is *cycle-consistent* if it is 1-consistent and every path from z to z connects a to a, for every variable z and every $a \in D_z$.

Definition 5.8 (Linked, fragmented, irreducible). $\mathcal{I}$ is *linked* if for every $z \in \mathrm{Var}(\mathcal{I})$ and every $a, b \in D_z$ some path from z to z connects a and b. $\mathcal{I}$ is *fragmented* if $\mathrm{Var}(\mathcal{I})$ splits into two disjoint nonempty sets X_1, X_2 with $\mathrm{Var}(C) \subseteq X_1$ or $\mathrm{Var}(C) \subseteq X_2$ for every C. $\mathcal{I}$ is *irreducible* if there is no instance $\mathcal{I}'$ with

(i) $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$;
(ii) every constraint of $\mathcal{I}'$ is a projection of a constraint of $\mathcal{I}$ onto some of its variables;
(iii) $\mathcal{I}'$ is not fragmented;
(iv) $\mathcal{I}'$ is not linked;
(v) the solution set of $\mathcal{I}'$ is not subdirect.

Formalization note 5.9 (Irreducibility is a Π_1 condition over a large family). Conditions (i)–(v) quantify over *all* instances built from projections of constraints of $\mathcal{I}$ — a finite but exponentially large family (any multiset of projections of any constraints onto any subsets of their variables). It is finite, so the predicate is decidable, but it is not something to unfold. Every use in §7 goes through Lemma 7.12 and *never* through the definition. Formalize it once, prove the two consequences, and never unfold it again.

The typo in the source at this point ("$\mathrm{Var}(C) \subseteq X_1$ or $\mathrm{Var}(C) \subseteq X_1$") is corrected above to X_2.

5.4 Weakening and cruciality

Definition 5.10 (Weakening). A constraint $R_1(y_1, \ldots, y_t)$ is *weaker or equivalent to* $R_2(z_1, \ldots, z_s)$ if $\{y_i\} \subseteq \{z_j\}$ and $R_2(\bar{z})$ implies $R_1(\bar{y})$; it is *weaker* if in addition $R_1(\bar{y})$ does not imply $R_2(\bar{z})$. *Weakening* a constraint C in $\mathcal{I}$ means replacing C by all constraints weaker than it. An instance $\mathcal{I}'$ is a *weakening* of $\mathcal{I}$ if $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$ and every constraint of $\mathcal{I}'$ is weaker or equivalent to a constraint of $\mathcal{I}$.

Formalization note 5.11 ("Replace by all weaker constraints"). Read literally this is an infinite operation: there are $2^{|D|^t}$ relations weaker than a given one on a given scope, and one must also range over all sub-scopes. It is finite for a fixed instance, but the resulting instance is astronomically large, and no algorithm performs this operation — it appears only inside proofs, as a device for reaching a crucial instance (Remark 5.13).

Two consequences for a formalization. First, weakening must be defined as a *relation between instances* ("$\mathcal{I}'$ is a weakening of $\mathcal{I}$"), not as a function producing one. Second, the descending chain argument in Remark 5.13 needs a well-founded measure; the natural one is the total number of tuples in all constraint relations, which strictly decreases when a constraint is properly weakened, together with the fact that only finitely many scopes are available. State that measure explicitly — "we cannot weaken forever" occurs three times in §7 and is the kind of step that silently fails to be well-founded if the scope is allowed to grow.

Definition 5.12 (Crucial). Let $D^{(\top)}$ be a reduction for $\mathcal{I}$. A variable y_i of a constraint $R(y_1, \ldots, y_t)$ is *dummy* if R does not depend on its i-th coordinate. A constraint $C \in \mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has no dummy variables, $\mathcal{I}^{(\top)}$ has no solution, and weakening C in $\mathcal{I}$ yields an instance with a solution in $D^{(\top)}$. The instance $\mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has at least one constraint and all of its constraints are crucial in $D^{(\top)}$.

Remark 5.13 (Reaching a crucial instance). If $\mathcal{I}^{(\top)}$ has no solution, then iteratively replacing each constraint by all weaker constraints without dummy variables terminates in an instance crucial in $D^{(\top)}$. Every relation so introduced is still a subuniverse of the corresponding product of domains.

5.5 Coverings

Definition 5.14 (Expanded covering)**.** $\mathcal{I}'$ is an *expanded covering* of $\mathcal{I}$, written $\mathcal{I}' \in \mathrm{Exp}(\mathcal{I})$, if there is $S\colon \mathrm{Var}(\mathcal{I}') \to \mathrm{Var}(\mathcal{I})$ with

(i) $S(x) = x$ for $x \in \mathrm{Var}(\mathcal{I}) \cap \mathrm{Var}(\mathcal{I}')$;
(ii) $D_x = D_{S(x)}$ for every $x \in \mathrm{Var}(\mathcal{I}')$;
(iii) for every constraint $R(x_1, \ldots, x_m)$ of $\mathcal{I}'$, either $S(x_1), \ldots, S(x_m)$ are pairwise distinct and $R(S(x_1), \ldots, S(x_m))$ is weaker or equivalent to a constraint of $\mathcal{I}$, or $S(x_1) = \cdots = S(x_m)$ and $\{(a, \ldots, a) : a \in D_{x_1}\} \subseteq R$.

It is a *covering* if moreover $R(S(x_1), \ldots, S(x_m)) \in \mathcal{I}$ for every constraint; a *tree-covering* if it is a covering and a tree-instance. $S(x)$ is the *parent* of x, and x a *child*.

Proposition 5.15 (Basic properties)**.** *Let $\mathcal{I}'$ be an expanded covering of $\mathcal{I}$.*

(a) *Replacing every x by $S(x)$ and deleting the constraints $R(x, \ldots, x)$ yields a weakening of $\mathcal{I}$.*
(b) *A weakening is an expanded covering with $S = \mathrm{id}$.*
(c) *Every solution of $\mathcal{I}$ lifts to a solution of $\mathcal{I}'$.*
(d) *If $\mathcal{I}$ is 1-consistent and $\mathcal{I}'$ is a tree-covering, the solution set of $\mathcal{I}'$ is subdirect.*
(e) *The union of two expanded coverings is an expanded covering.*
(f) *An expanded covering of an expanded covering is an expanded covering.*
(g) *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*
(h) *Every reduction of $\mathcal{I}$ extends to $\mathcal{I}'$, and a 1-consistent one extends to a 1-consistent one.*

Formalization note 5.16 (Variable renaming is the hidden cost)**.** Expanded coverings are where a formalization pays. The paper writes "we rename the variables so that the only common variable of Υ_x and $\mathcal{J}$ is x", and forms conjunctions $\bigwedge_{\mathcal{J}\in\Omega} \bot(\mathcal{J})$ of instances whose variable sets must first be made disjoint. Doing this with a concrete variable type means a fresh-name supply and a renaming operation, with a lemma that renaming preserves each of 1-consistency, cycle-consistency, cruciality, and being a covering — eight lemmas that the paper does not state.

The cheaper design is to make the variable type a parameter and build instances by *disjoint union* of variable types: Υ_x and $\mathcal{J}$ are placed over $V_1 \oplus V_2$ quotiented by identifying the single shared variable. Renaming then disappears into the coproduct, and the eight lemmas become transport along an equivalence. This is the single most important representation decision in §5, and it should be made before anything downstream is written.

5.6 Induced congruences and connectedness

Definition 5.17 (Con_1)**.** For R of arity m and $i \in [m]$, $\mathrm{Con}_1(R, i)$ is the binary relation

$$\{(y, y') \;:\; \exists\, \bar{x} \;\; R(\bar{x}[i \mapsto y]) \wedge R(\bar{x}[i \mapsto y'])\}.$$

For a constraint $C = R(x_1, \ldots, x_m)$ we write $\mathrm{Con}_1(C, x_i)$ for $\mathrm{Con}_1(R, i)$. For an instance, $\mathrm{Con}(\mathcal{I}, x) = \{\mathrm{Con}_1(C, x) : C \in \mathcal{I}\}$ and $\mathrm{Con}(\mathcal{I}) = \bigcup_x \mathrm{Con}(\mathcal{I}, x)$.

Lemma 5.18. *The i-th variable of R is rectangular if and only if R is stable under $\mathrm{Con}_1(R, i)$. If R is subdirect and its i-th variable is rectangular, then $\mathrm{Con}_1(R, i)$ is a congruence.*

Formalization note 5.19. $\mathrm{Con}_1(R, i)$ is always reflexive and symmetric but is *not* transitive in general, hence not always a congruence; Lemma 5.18 is what makes it one, and its hypotheses (subdirect *and* rectangular at i) are both needed. The notation Con_1 invites the reader to assume congruence-hood; a formalization should give the raw relation a neutral name and reserve Con_1 for the case where the lemma applies. Note also $\mathrm{Con}_1(C, x_i)$ is ill-defined if x occurs twice in the scope: it depends on the *position* i, not the variable.

Definition 5.20 (Types of relations and instances)**.** R is *of PC type* (resp. *linear type*) if R is rectangular and every $\mathrm{Con}_1(R, i)$ is a PC (resp. linear) congruence. An instance is of PC or linear type if all its constraints are.

Definition 5.21 (Adjacency, connectedness)**.** Two congruences σ_1, σ_2 on $\mathbf{D}_x$ are *adjacent* if there is a reflexive bridge from σ_1 to σ_2. Two rectangular constraints C_1, C_2 are *adjacent in a common variable x* if $\mathrm{Con}_1(C_1, x)$ and $\mathrm{Con}_1(C_2, x)$ are adjacent. An instance $\mathcal{I}$ is *connected* if all its constraints are rectangular, all congruences in $\mathrm{Con}(\mathcal{I})$ are irreducible, and the graph on the constraints with edges the adjacent pairs is connected.

Note that every proper congruence is adjacent to itself, via $\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4)$.

6 Properties of strong subuniverses

This section is Zhuk's §2.3. It is the *interface* between the algebraic theory and the CSP argument: §7 uses the results below and nothing else from §§4. Proofs are deferred to §5 of [10], which is not reproduced here; §12 budgets it.

Its shape is worth naming. Four of the five groups say that the relation $\lhd\lhd\lhd$ and the types T *propagate*: forward and backward along surjective homomorphisms, down to quotients, across products of a congruence, and through subdirect relations. The fifth — Theorem 6.13 — is the one genuinely hard statement, and it is what converts a failure of consistency into a bridge. Everything the CSP proof does with bridges traces back to it.

Convention 6.1. Where a type is not specified, T ranges over $\{\text{BA}, \text{C}, \text{S}, \text{PC}, \text{L}, \text{D}\}$. Where a multi-type $\mathcal{M}T$ appears, T ranges over $\{\text{PC}, \text{L}, \text{D}\}$.

6.1 Ubiquity

Lemma 6.2 (Ubiquity)**.** *If $B \lhd\lhd\lhd A$ and $|B| > 1$, then $C <_T^A B$ for some C and some $T \in \{\text{BA}, \text{C}, \text{L}, \text{PC}\}$.*

Lemma 6.2 is the formal content of Informal Claim 7.1: every domain of size at least 2 that has been reached by a chain of strong reductions admits another strong reduction. It is what guarantees the outer loop of the algorithm makes progress, and it is the reason the algorithm terminates with singleton domains or a linear instance.

Formalization note 6.3. The hypothesis is $B \lhd\lhd\lhd A$, not $B \le A$. A subalgebra reached by no chain of strong reductions need admit nothing. In the algorithm this hypothesis is maintained as an invariant of the reduction loop; in a formalization it should be carried in the state, not re-derived.

6.2 Propagation

Lemma 6.4 (Propagation along a homomorphism)**.** Let $f\colon \mathbf{A} \to \mathbf{A}'$ be a surjective homomorphism. Then

(f) $C \lhd\lhd\lhd^A B \Rightarrow f(C) \lhd\lhd\lhd f(B)$;
(b) $C' \lhd\lhd\lhd^{A'} B' \Rightarrow f^{-1}(C') \lhd\lhd\lhd f^{-1}(B')$;
(ft) $C <_{T(\sigma)}^A B \lhd\lhd\lhd A \Rightarrow (f(C) = f(B)$ or $f(C) <_{\text{S}} f(B)$ or $f(C) <_T^{A'} f(B))$;
(bt) $C' <_{T(\sigma)}^{A'} B' \Rightarrow f^{-1}(C') <_{T(f^{-1}(\sigma))}^A f^{-1}(B')$;
(fs) $T \in \{\text{BA}, \text{C}, \text{S}\}$ and $C <_T B \Rightarrow f(C) \le_T f(B)$;
(fm) $C \le_{\mathcal{M}T}^A B \lhd\lhd\lhd A$ and $f(B)$ S-free $\Rightarrow f(C) \le_{\mathcal{M}T}^{A'} f(B)$;
(bm) $C' \le_{\mathcal{M}T}^{A'} B' \lhd\lhd\lhd A' \Rightarrow f^{-1}(C') \le_{\mathcal{M}T}^A f^{-1}(B')$.

Formalization note 6.5 (The three-way disjunction in (ft))**.** Clause (ft) is the characteristic shape of this theory and the reason it is usable. Pushing a strong subset forward along a homomorphism can

- collapse it ($f(C) = f(B)$: the reduction becomes vacuous),
- degrade it ($f(C) <_{\text{S}} f(B)$: still strong, but the type is lost),
- or preserve it ($f(C) <_T^{A'} f(B)$: same type).

The type S exists to be the value of the middle case. Every downstream proof that pushes a reduction through a quotient does a three-way case split here, and the middle case is the one that is easy to forget; in [9] it is the source of the "go forward and backward from global to local" induction that this paper eliminates.

Note the asymmetry: the backward clauses (b), (bt), (bm) are clean implications with no disjunction. Preimages behave; images do not.

Corollary 6.6 (Propagation to a quotient)**.** *Let δ be a congruence on* $\mathbf{A}$*. Then* (f) $C \lhd\lhd\lhd^A B \Rightarrow C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta$*;* (t) $C <^A_{T(\sigma)} B \lhd\lhd\lhd A \Rightarrow (C/\delta = B/\delta$ *or* $C/\delta <_{\textsc{s}} B/\delta$ *or* $C/\delta <^{A/\delta}_T B/\delta)$*;* (s) *for* $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$*,* $C <_T B \Rightarrow C/\delta \leq_T B/\delta$*;* (m) $C \leq^A_{\mathcal{M}T} B \lhd\lhd\lhd A$ *and* B/δ s*-free* $\Rightarrow C/\delta \leq^{A/\delta}_{\mathcal{M}T} B/\delta$*.*

Corollary 6.7 (Reflection from a quotient)**.** *Let δ be a congruence on* $\mathbf{A}$ *and* $B, C \leq \mathbf{A}$*. Then* $C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta \iff C \circ \delta \lhd\lhd\lhd^A B \circ \delta$*, and* $C/\delta <^{A/\delta}_T B/\delta \iff C \circ \delta <^A_T B \circ \delta$*.*

Corollary 6.8 (Multiplying by a congruence)**.** *Let δ be a congruence on* $\mathbf{A}$*. Then* (f) $C \lhd\lhd\lhd^A B \Rightarrow C \circ \delta \lhd\lhd\lhd^A B \circ \delta$*;* (t) $C <^A_{T(\sigma)} B \lhd\lhd\lhd^A A \Rightarrow (C \circ \delta = B \circ \delta$ *or* $C \circ \delta <^A_{\textsc{s}} B \circ \delta$ *or* $C \circ \delta <^A_T B \circ \delta)$*;* (e) *if* $\delta \subseteq \sigma$ *and* $C <^A_{T(\sigma)} B \lhd\lhd\lhd^A A$ *then* $C \circ \delta <^A_T B \circ \delta$*.*

Corollary 6.9 (Propagation to relations)**.** Let $R \leq_{\text{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and $B_i \lhd\lhd\lhd A_i$. Then

(r) $R \cap (B_1 \times \cdots \times B_m) \dot{\lhd}\dot{\lhd}\dot{\lhd} R$;
(r1) $\text{pr}_1(R \cap (B_1 \times \cdots \times B_m)) \dot{\lhd}\dot{\lhd}\dot{\lhd} A_1$;
(b) if $C_i \lhd\lhd\lhd^{A_i} B_i$ for all i then $R \cap (C_1 \times \cdots \times C_m) \dot{\lhd}\dot{\lhd}\dot{\lhd}^R R \cap (B_1 \times \cdots \times B_m)$;
(b1) under the same hypothesis, $\text{pr}_1(R \cap \prod C_i) \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A_1} \text{pr}_1(R \cap \prod B_i)$;
(m) if $C_i \leq^{A_i}_{\mathcal{M}T} B_i$ for all i then $R \cap \prod C_i \dot{\leq}^R_{\mathcal{M}T} R \cap \prod B_i$;
(m1) if in addition $\text{pr}_1(R \cap \prod B_i)$ is s-free, then $\text{pr}_1(R \cap \prod C_i) \dot{\leq}^{A_1}_{\mathcal{M}T} \text{pr}_1(R \cap \prod B_i)$.

For the two absorption types the conclusion is stronger — no chain, no dot on the type, and no subdirectness hypothesis:

Lemma 6.10 ([11, Cor. 6.1.2, 6.9.2])**.** *Let* $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ *with* $\text{pr}_1(R) = A_1$*, and let* $C_i \leq_T A_i$ *for every i, where* $T \in \{\textsc{ba}, \textsc{c}\}$ *— and, when* $T = \textsc{ba}$*, with a* single *binary term t witnessing all m absorptions. Then* $\text{pr}_1(R \cap (C_1 \times \cdots \times C_m)) \dot{\leq}_T A_1$*, again with the same t when* $T = \textsc{ba}$*.*

Formalization note 6.11 (The source's restatement is false; this is the repair)**.** Lemma 6.10 is the workhorse of case (2) of Theorem 7.1, and [10, Lemma 19] restates it from [11] *dropping two hypotheses*: the subdirectness $\text{pr}_1(R) = A_1$, and — in the ba case — that the C_i absorb with a common term. Both are present in the cited source ([11, Cor. 6.1.2] reads "Suppose $R \leq A_1 \times \cdots \times A_n$, $\text{pr}_1(R) = A_1$, $B_i \leq_{\textsc{ba}(t)} A_i$ for every i"), and without the first the statement is false.

Counterexample to the printed form. Take $R = \{(0, 0)\} \leq \mathbf{Z}_p \times \mathbf{Z}_p$ — a subuniverse, by idempotence — and $C_i = A_i$, which is allowed because $\leq_T$ admits equality. Then $R \cap (C_1 \times C_2) = R$ and $\text{pr}_1(R) = \{0\}$, which is neither empty nor all of $\mathbf{Z}_p$, so the conclusion asserts $\{0\} <_{\textsc{ba}} \mathbf{Z}_p$ — contradicting Lemma 6.24, the paper's own Lemma 29. The failure is exactly the missing subdirectness: $\text{pr}_1(R) = \{0\} \neq \mathbf{Z}_p$.

The repair is to restore both hypotheses, as above. A formalization must also check that they hold at each of the call sites — in particular in the proof of Theorem 7.1(2), where the lemma is applied to the relation computing E_2 from E_1. The common-term condition is the reason [11] writes $\leq_{\textsc{ba}(t)}$ with the term in the notation, and carrying t in the type is the cheapest way to keep it.

This is the single highest-value import to formalize first: small, self-contained, used everywhere, and adjacent to what the predecessor project already proves.

6.3 Intersections

Lemma 6.12 (Intersecting with a strong subset)**.** *If $B \lhd\lhd\lhd A$ and $D \lhd\lhd\lhd A$ then* (i) $B \cap D \;\dot{\lhd}\dot{\lhd}\dot{\lhd}\; A$, *and* (t) $C <^{A}_{T(\sigma)} B \Rightarrow C \cap D \,\dot{\leq}^{A}_{T(\sigma)} B \cap D$.

The next theorem is the heart of §6. Read it as follows: if a family of strong reductions is *jointly* inconsistent but *minimally* so — dropping any one of them restores consistency — then the types must agree, and in the linear case one gets bridges between the witnessing congruences. Bridges are produced nowhere else.

Theorem 6.13 (Stable intersection)**.** Suppose

(i) $C_i <^{A}_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{l}, \textsc{pc}\}$, for $i \in [m]$, $m \geq 2$;
(ii) $\bigcap_{i\in[m]} C_i = \varnothing$;
(iii) $B_j \cap \bigcap_{i\neq j} C_i \neq \varnothing$ for every $j \in [m]$.

Then one of:

(ba) $T_1 = \cdots = T_m = \textsc{ba}$;
(l) $T_1 = \cdots = T_m = \textsc{l}$, and for all $k, \ell \in [m]$ there is a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \sigma_\ell$;
(c) $m = 2$ and $T_1 = T_2 = \textsc{c}$;
(pc) $m = 2$, $T_1 = T_2 = \textsc{pc}$ and $\sigma_1 = \sigma_2$.

Corollary 6.14 (Stable intersection, relational form)**.** Suppose $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$, $C_i <^{A_i}_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A_i$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{l}, \textsc{pc}\}$ and $m \geq 2$, $R \cap (C_1 \times \cdots \times C_m) = \varnothing$, and $R \cap (C_1 \times \cdots \times B_j \times \cdots \times C_m) \neq \varnothing$ for every j. Then one of: (ba) all $T_i = \textsc{ba}$; (l) all $T_i = \textsc{l}$ and for all k, ℓ a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$; (c) $m = 2$ and $T_1 = T_2 = \textsc{c}$; (pc) $m = 2$, $T_1 = T_2 = \textsc{pc}$, $\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$, and $\{(a/\sigma_1, b/\sigma_2) : (a, b) \in R\}$ is bijective.

Remark 6.15. Several applications need more than one restriction on a single coordinate of R. The statement is kept simple; to apply it, duplicate the coordinate (form $R' = \{(\bar{a}, a_k) : \bar{a} \in R\}$) and restrict the copies separately.

Formalization note 6.16 (Minimality is a hypothesis, not a construction)**.** Hypotheses (ii)–(iii) of Theorem 6.13 say the family is *critically* inconsistent. In applications one starts with an inconsistent family and shrinks it to a critical subfamily. That shrinking step is left implicit in the paper ("let $\mathcal{M}$ be a minimal set of children of z we need to restrict"). It is easy — a minimal subset of a finite set with a monotone property — but it must be stated as a lemma, because it is performed at least four times and each time the property being minimised is slightly different. State once: *if P is a monotone predicate on subsets of a finite set X and $P(X)$ holds, there is $M \subseteq X$ minimal with $P(M)$*, and instantiate.

6.4 Multi-types

Lemma 6.17. *If $C \leq^{A}_{\mathcal{M}T} B$ then $C <^{A}_{T} \ldots <^{A}_{T} B$ and $C \lhd\lhd\lhd^{A} B$.*

That is: an intersection of type-T subsets, though not itself obtained by a single reduction of type T, is reachable by a *chain* of type-T reductions. This is what makes $\mathcal{M}T$ compatible with $\lhd\lhd\lhd$, and it is used whenever the algorithm reduces several variables at once.

Lemma 6.18 (Preserving linkedness)**.** *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, $C_i \leq^{A_i}_{\mathcal{M}\mathrm{D}} B_i \lhd\lhd\lhd A_i$ for $i = 1, 2$, let S be the rectangular closure of R, and suppose $R \cap (B_1 \times B_2) \neq \varnothing$ and $S \cap (C_1 \times C_2) \neq \varnothing$. Then $R \cap (C_1 \times C_2) \neq \varnothing$.*

Lemma 6.19 (Maximal multi-type extension)**.** *Let $C_1 <^A_{\mathcal{M}T} B_1 \lhd\lhd\lhd A$, $B_2 \lhd\lhd\lhd A$, $C_1 \cap B_2 = \varnothing$, $B_1 \cap B_2 \neq \varnothing$, and let σ be a maximal congruence on $\mathbf{A}$ with $(C_1 \circ \sigma) \cap B_2 = \varnothing$. Then $\sigma = \omega_1 \cap \cdots \cap \omega_s$ where each ω_i is a congruence of type T on $\mathbf{A}$ with $\omega_i^* \supseteq B_1^2$.*

6.5 Auxiliary imports

The following are used but not proved in [10].

Lemma 6.20 ([6, Lem. 4.7])**.** *If w is an idempotent WNU operation of arity n on a finite set A, then* $\mathrm{Clo}(w)$ *contains a special idempotent WNU operation of arity $n^{n!}$.*

Lemma 6.21 ([9, Cor. 8.17.1])**.** *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta} = A^2$, then σ is a perfect linear congruence.*

Lemma 6.22. *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta}$ linked, then σ is a perfect linear congruence.*

Proof. Let $\delta^{-1}(y_1, y_2, x_1, x_2) = \delta(x_1, x_2, y_1, y_2)$, which is again a bridge from σ to σ. Since $\widetilde{\delta}$ is linked, $(\widetilde{\delta} \circ \widetilde{\delta^{-1}})^N = A^2$ for N large. Composing $2N$ bridges $\delta, \delta^{-1}, \ldots$ by Lemma 3.26 gives a bridge δ' from σ to σ with $\widetilde{\delta'} = A^2$; apply Lemma 6.21. □

Formalization note 6.23 ("for sufficiently large N")**.** The step $(\widetilde{\delta} \circ \widetilde{\delta^{-1}})^N = A^2$ deserves a lemma of its own: if a reflexive symmetric relation ρ on a finite set A has connected graph, then $\rho^N = A^2$ for $N \geq |A|$. Reflexivity is what makes the powers increase monotonically; $\widetilde{\delta}$ is reflexive because $(a, a, a, a) \in \delta$ is not assumed — so the reflexivity must come from δ being a bridge from σ to σ with σ reflexive, via $\mathrm{pr}_{1,2}(\delta) \supsetneq \sigma$. Check this: it is the sort of step where the informal proof is right for a reason it does not give.

Lemma 6.24. *If $B \leq \mathbf{Z}_p \times \cdots \times \mathbf{Z}_p$ then there is no $C <_T B$ with $T \in \{\textsc{ba}, \textsc{c}\}$.*

Proof. It suffices to check that for every term τ, if the last variable of $\tau^{\mathbf{Z}_p}(a, \ldots, a, x)$ is not dummy then it takes all values; hence no absorbing subuniverse exists. □

7 The main statements

This section is Zhuk's §§3.3–3.4. Everything converges here. Theorem 7.1 is the single induction that carries the whole proof; the three theorems the algorithm consumes are corollaries of it.

7.1 The three claims the algorithm needs

The algorithm of §8 is justified by exactly three facts. Informally:

(IC1) every domain of size ≥ 2 has a strong subset, or an equivalence σ with $D_x/\sigma \cong \mathbb{Z}_p$;
(IC2) reducing a variable's domain to a strong subset of it does not destroy solvability, for a consistent enough instance;
(IC3) for a linked, consistent enough, strong-subset-free instance, the set of "linear cosets" containing a solution is empty, full, or a codimension-one affine subspace.

(IC1) is Lemma 6.2 together with Lemma 4.14. (IC2) is Theorem 7.6. (IC3) is Theorem 7.8.

7.2 The main induction

Theorem 7.1 (Main inductive claim)**.** *Let $\mathcal{I}$ be an irreducible cycle-consistent instance and let $D^{(1)}$ be a 1-consistent reduction for $\mathcal{I}$ with $D^{(1)} \lhd\lhd\lhd D$.*

(1) If $\mathcal{I}$ is crucial in $D^{(1)}$ then (1a) holds, and (1b) or (1c) holds:
   (1a) every constraint of $\mathcal{I}$ has the parallelogram property;
   (1b) $\mathcal{I}$ is a connected linear-type instance whose solution set is subdirect;
   (1c) there is an expanded covering $\mathcal{J}$ of $\mathcal{I}$, crucial in $D^{(1)}$, with a linked connected subinstance Υ whose solution set is not subdirect.

(2) If $D^{(2)} \leq_T D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$ with $T \in \{\text{BA}, \text{C}\}$, and $\mathcal{I}^{(1)}$ has a solution, then $\mathcal{I}^{(2)}$ has a solution.

Formalization note 7.2 (The induction, stated precisely)**.** The proof reads "we prove the claim by induction on the size of $D^{(1)}$", and this one line hides the entire well-foundedness argument. Making it precise is the first thing a formalization must do, because *the two halves of the conclusion are proved by mutual induction and each invokes the other at a smaller reduction.*

The measure is

$$\mu(D^{(1)}) = \sum_{x \in \mathrm{Var}(\mathcal{I})} |D_x^{(1)}| \in \mathbb{N},$$

and the statement being proved by strong induction on μ is the conjunction of (1) and (2), *universally quantified over $\mathcal{I}$*:

$$P(k): \quad \forall \mathcal{I},\ \forall D^{(1)} \text{ with } \mu(D^{(1)}) \leq k,\ \text{(1) and (2) hold.}$$

The quantifier over $\mathcal{I}$ must be *inside*, because every use of the inductive hypothesis changes the instance: (2) applies it to a weakening $\mathcal{I}'$ of $\mathcal{I}^{(2)}$; Case 1 of (1) applies it to $\mathcal{I}$ with $D^{(2)}$; Case 2 applies it to weakenings and to expanded coverings. An induction that fixed $\mathcal{I}$ and inducted on $D^{(1)}$ alone does not go through.

Two further points that the one-line phrasing conceals.

(a) *The instance grows.* Expanded coverings have *more* variables than $\mathcal{I}$. So μ is not monotone in the instance, and the induction is genuinely on $\mu(D^{(1)})$ alone — with the reduction of a covering being the extension of $D^{(1)}$ along the parent map (Proposition 5.15(h)), whose μ is *larger*. Every appeal to the inductive hypothesis at a covering must therefore be checked to occur at a strictly smaller reduction, not merely at a different instance. In Case 2 of (1) the reductions used are the $D^{(B,\perp)}$, which are contained in $D^{(1)}$ restricted at z to a proper subset B; that is where the decrease comes from.

(b) *Which appeals decrease.* (2) invokes the hypothesis at $D^{(2)} \subsetneq D^{(1)}$: strict because $D^{(2)} <_T D^{(1)}$ has $C \lneq B$ at some variable. Case 1 of (1) likewise. Case 2 invokes it at $D^{(B,\perp)} \subseteq D^{(B,\top)}$, which is $D^{(1)}$ with the z-domain cut to $B \lneq D^{(1)}_z$. In every case the decrease is at a single variable and by at least one element. A formalization should extract this as a lemma: *if $D' \leq_T D$ with T any type and $D' \neq D$, then $\mu(D') < \mu(D)$*, and use it uniformly.

Proof of (2). Suppose $\mathcal{I}^{(2)}$ has no solution. Weaken $\mathcal{I}^{(2)}$ to an instance $\mathcal{I}'$ crucial in $D^{(2)}$ (Remark 5.13). By the inductive hypothesis applied to $\mathcal{I}'$ and $D^{(2)}$, $\mathcal{I}'$ satisfies (1a) and (1b) or (1c).

Case (1c). There is an expanded covering $\mathcal{J}$ of $\mathcal{I}'$, crucial in $D^{(2)}$, with a linked connected subinstance Υ. Let x be a variable of a constraint $C \in \Upsilon$. By Lemma 7.14(p), $\mathrm{Con}_1(C, x)$ is a perfect linear congruence; choose $\zeta \subseteq \mathbf{D}_x \times \mathbf{D}_x \times \mathbf{Z}_p$ with $(y_1, y_2, 0) \in \zeta \iff (y_1, y_2) \in \mathrm{Con}_1(C, x)$ and $\mathrm{pr}_{1,2}(\zeta) = \mathrm{Con}_1(C, x)^*$.

Replace the variable x of C in $\mathcal{J}$ by a fresh x' and add the constraint $\zeta(x, x', z)$ with z fresh of domain $\mathbf{Z}_p$; call the result Θ, and extend $D^{(1)}, D^{(2)}$ by $D^{(1)}_{x'} = D^{(2)}_{x'} = D_{x'}$. Let E_i be the set of $b \in \mathbf{Z}_p$ such that Θ has a solution in $D^{(i)}$ with $z = b$. By Lemma 6.10, $E_2 \dot{\leq}_T E_1$.

Since $\mathcal{J}$ is crucial in $D^{(2)}$, $\Theta^{(2)}$ has a solution but none with $z = 0$; since $\mathcal{I}^{(1)}$ has a solution, so does $\mathcal{J}^{(1)}$, hence $\Theta^{(1)}$ has one with $z = 0$. Thus $0 \in E_1$, $0 \notin E_2$, and E_1 has at least two elements. As $\mathbf{Z}_p$ has no proper subalgebra of size > 1, $E_1 = \mathbf{Z}_p$; then $E_2 <_T \mathbf{Z}_p$ with $T \in \{\textsc{ba}, \textsc{c}\}$, contradicting Lemma 6.24.

Case (1b). Corollary 7.21 gives a contradiction directly. □

Formalization note 7.3 (Three gaps in the case (1c) argument). (a) "E_1 contains at least two different elements, $0 \in E_1$ and $0 \notin E_2$, hence $E_1 = \mathbf{Z}_p$" uses that E_1 is a subuniverse of $\mathbf{Z}_p$ — true because $E_1 = \Theta^{(1)}(z)$ is a relation defined by an instance (Definition 5.3) — *and* that the subuniverses of $\mathbf{Z}_p$ are the singletons and the whole set. The latter needs $w^{\mathbf{Z}_p}$ to generate: from $a \neq b$ one reaches every element. Prove it; it is one line but it is the crux.

(b) $E_2 \dot{\leq}_T E_1$ is applied with the *dotted* conclusion of Lemma 6.10, and the argument then treats E_2 as a proper subuniverse. The case $E_2 = \varnothing$ must be handled: it is excluded because $\Theta^{(2)}$ has a solution. Say so.

(c) The choice of ζ is by Definition 3.28, which supplies ζ with $\mathrm{pr}_{1,2}\,\zeta = \sigma^*$ and $(a_1, a_2, b) \in \zeta \Rightarrow ((a_1, a_2) \in \sigma \iff b = 0)$. The proof uses the *biconditional* $(y_1, y_2, 0) \in \zeta \iff (y_1, y_2) \in \mathrm{Con}_1(C, x)$, which is stronger: it needs, for each $(y_1, y_2) \in \sigma$, that $(y_1, y_2, 0) \in \zeta$. This follows from $\mathrm{pr}_{1,2}\,\zeta = \sigma^* \supseteq \sigma$ plus the implication, but the step is suppressed.

Proof of (1), *sketch of the structure.* First, $|D^{(1)}_x| > 1$ for some x: otherwise $\mathcal{I}^{(1)}$ has all domains singletons and 1-consistency makes it solvable, contradicting cruciality.

Case 1: some $D^{(1)}_x$ has a nontrivial BA *or central subuniverse.* Lemma 7.17 yields a 1-consistent reduction $D^{(2)} \leq_T D^{(1)}$, $T \in \{\textsc{ba}, \textsc{c}\}$. One checks $\mathcal{I}$ is crucial in $D^{(2)}$: for a weakening $\mathcal{J}$ of a constraint, $\mathcal{J}^{(1)}$ has a solution, so by the inductive hypothesis (part (2) at $D^{(2)}$) $\mathcal{J}^{(2)}$ has one. Applying the inductive hypothesis (part (1)) at $D^{(2)}$ gives the conclusion.

Case 2: otherwise. Lemma 6.2 gives $E <^{D_z}_{T(\sigma)} D^{(1)}_z$ for some z and $T \in \{\textsc{pc}, \textsc{l}\}$. Property (1a) is obtained by choosing, for each x, a minimal $D^{(2)}_x \leq_{\mathcal{M}T} D^{(1)}_x$ containing the value at x of a solution produced by weakening a fixed constraint C; Lemma 6.17 gives $D^{(2)} \lhd\!\lhd\!\lhd^D D^{(1)}$

and Lemma 7.11 splits into two subcases, in one of which the inductive hypothesis applies and in the other of which Lemma 7.18 applies directly.

For (1b)/(1c) one forms, for each B in the family $\mathcal{B} = \{B : B <_{T(\sigma)}^{D_z} D_z^{(1)}\}$, the reduction $D^{(B,\top)}$ that equals $D^{(1)}$ off z and B at z, and the maximal 1-consistent reduction $D^{(B,\perp)} \subseteq D^{(B,\top)}$ (possibly empty). Lemma 7.16 supplies tree-coverings $\Upsilon_{B,x}$ computing $D_x^{(B,\perp)}$, and their conjunction Υ_x works uniformly in B. Writing $\mathcal{B}_0$ for the B with $D^{(B,\perp)}$ nonempty, the argument splits:

- $\mathcal{B}_0 = \varnothing$: the type must be L, and Lemma 7.19 applied to a suitably weakened tree-covering produces, for any two constraints sharing a variable, a reflexive bridge between their Con_1's. Hence $\mathcal{I}$ is connected, and (1b) or (1c) holds according as its solution set is subdirect.
- $\mathcal{B}_0 \neq \varnothing$: one constructs a family Ω of weakenings of $\mathcal{I}$ that is *critically* unable to solve all of $\mathcal{B}_0$ (conditions 1–4 of Construction 7.5) and argues by whether some member fails (1b).

□

Formalization note 7.4 (Construction 7.5 is the hardest object)**.** The set Ω is specified by four conditions, one of which (condition 3) is a minimality condition "if we replace any instance in Ω by all weaker instances then 2 is not satisfied". The paper obtains Ω by starting from $\{\mathcal{I}\}$ and repeatedly weakening, justified by "we cannot weaken forever". Condition 4 is then *derived* from condition 3.

For a formalization this is the single most demanding step in §7, for three reasons. (i) The iteration weakens a *set* of instances, so the termination measure is a measure on finite sets of instances, not on instances — see Hazard 5.11. (ii) The derivation of condition 4 from condition 3 is a two-line argument that instantiates condition 3 at $\Omega \cup \{\mathcal{J}_C : C \in \mathcal{J}\} \setminus \{\mathcal{J}\}$ and reads off an existential; the existential witness B depends on $\mathcal{J}$, so condition 4 is a $\Pi\Sigma$ statement and must be stated as such. (iii) The subsequent Subcase 1 builds a *second* descending sequence $\mathcal{B}_1 \supseteq \mathcal{B}_2 \supseteq \dots$ inside the first, again terminating by finiteness. Nested well-founded constructions of this shape are where informal proofs most often hide a circularity, and this one should be isolated and stated as a standalone lemma before Theorem 7.1 is attempted.

Remark 7.5 (Construction of Ω)**.** Ω is a finite set of weakenings of $\mathcal{I}$ with:

1. every member is a weakening of $\mathcal{I}$;
2. $\bigcap_{\mathcal{J} \in \Omega} \mathrm{Sol}(\mathcal{J}) = \varnothing$, where $\mathrm{Sol}(\mathcal{J}) = \{B \in \mathcal{B}_0 : \mathcal{J}^{(B,\perp)} \text{ has a solution}\}$;
3. replacing any member by all weaker instances destroys 2;
4. for every $\mathcal{J} \in \Omega$ there is $B \in \mathcal{B}_0$ with (a) $\mathcal{J}$ crucial in $D^{(B,\perp)}$ and (b) $B \in \mathrm{Sol}(\mathcal{J}')$ for every $\mathcal{J}' \in \Omega \setminus \{\mathcal{J}\}$.

7.3 The three consumed theorems

Theorem 7.6 (Reductions are safe)**.** *Let Θ be a cycle-consistent irreducible instance and $B <_T^{D_x} D_x$ with $T \in \{\text{BA}, \text{C}, \text{PC}\}$. Then Θ has a solution if and only if Θ has a solution with $x \in B$.*

Proof. For $T = \text{PC}$ this is Theorem 7.7. For $T \in \{\text{BA}, \text{C}\}$, Lemma 7.17 gives a 1-consistent reduction $D^{(1)} \leq_T D$ with $D_x^{(1)} \subseteq B$; by Theorem 7.1(2), $\Theta^{(1)}$ has a solution, hence Θ has one with $x \in B$. □

Theorem 7.7 (PC reductions do not kill all solutions)**.** *If $\mathcal{I}$ is cycle-consistent and irreducible, $B <_{\text{PC}(\sigma)}^{D_y} D_y$ for some $y \in \mathrm{Var}(\mathcal{I})$, and $\mathcal{I}$ has a solution, then $\mathcal{I}$ has a solution with $y \in B$.*

Theorem 7.8 (Codimension one)**.** Suppose

(i) $\mathcal{I}$ is a linked cycle-consistent irreducible instance with $\mathrm{Var}(\mathcal{I}) = \{x_1, \dots, x_m\}$;
(ii) $\mathbf{D}_{x_i}$ is S-free for every i;
(iii) weakening all constraints of $\mathcal{I}$ gives an instance with subdirect solution set;
(iv) σ_{x_i} is the intersection of all linear congruences σ on $\mathbf{D}_{x_i}$ with $\sigma^* = D^2_{x_i}$;
(v) $L_{x_i} = \mathbf{D}_{x_i}/\sigma_{x_i}$;
(vi) $\phi\colon \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \to L_{x_1} \times \cdots \times L_{x_m}$ is a homomorphism, q_i prime;
(vii) weakening any one constraint of $\mathcal{I}$ gives an instance with a solution in $\phi(\bar{a})$ for every $\bar{a} \in \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k}$.

Then $\Delta = \{\bar{a} : \mathcal{I} \text{ has a solution in } \phi(\bar{a})\}$ is empty, full, or an affine subspace of codimension 1 — the solution set of a single linear equation.

Proof. If Δ is full we are done. Otherwise pick $\bar{b} \notin \Delta$ and regard $\phi(\bar{b})$ as a reduction $D^{(1)}$ for $\mathcal{I}$; condition (vii) says $\mathcal{I}$ is crucial in $D^{(1)}$.

Step 1: some $\mathrm{Con}_1(C, x)$ *is a perfect linear congruence.* By Lemma 7.11, either $C_0^{(1)} = \varnothing$ for some constraint C_0, or $D^{(1)}$ is 1-consistent.

- If $C_0^{(1)} = \varnothing$: cruciality forces $\mathcal{I} = \{C_0\}$, say $C_0 = R(y_1, \dots, y_t)$. By Lemma 7.18, R has the parallelogram property and $\mathrm{Con}_1(R, 1)$ is linear with $\mathrm{Con}_1(R, 1)^* = D^2_{y_1}$. By Lemma 4.14, $\mathbf{D}_{y_1}/\delta \cong \mathbf{Z}_p$; with ψ the quotient map, $\zeta = \{(a_1, a_2, b) : \psi(a_1) - \psi(a_2) = b\}$ witnesses perfection.
- If $D^{(1)}$ is 1-consistent: by Theorem 7.1(1) every constraint has the parallelogram property and (1b) or (1c) holds. In either case Lemma 7.14(p) applies — under (1c) to the linked connected subinstance, whose containing a constraint of $\mathcal{I}$ is forced by condition (iii); under (1b) to $\mathcal{I}$ itself.

Step 2: add a $\mathbf{Z}_p$ *coordinate.* Let ζ witness perfection of $\mathrm{Con}_1(C, x)$. Add a variable z with domain $\mathbf{Z}_p$, replace x in C by a fresh x', and add $\zeta(x, x', z)$, giving $\mathcal{I}'$. Put

$$L = \{(\bar{a}, b) : \mathcal{I}' \text{ has a solution with } z = b \text{ in } \phi(\bar{a})\} \le \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \times \mathbf{Z}_p.$$

By condition (vii), $\mathrm{pr}_{1..k}(L)$ is full; and $(\bar{b}, 0) \notin L$. So L has dimension k and is the solution set of one linear equation. If that equation is $z = c$ with $c \neq 0$ then $\Delta = \varnothing$; otherwise setting $z = 0$ gives an equation defining Δ, of dimension $k - 1$. $\square$

Formalization note 7.9 (The linear algebra of Step 2)**.** Step 2 uses four facts about subgroups of $\mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \times \mathbf{Z}_p$ that are stated as if obvious:

(a) L is a subuniverse, hence (the q_i, p being prime and w being a sum) a subgroup;
(b) "dimension" is well defined for a subgroup of a product of $\mathbf{Z}_q$ with *different* primes q. It is not a vector space; the correct statement is that such a group splits as a product over primes of $\mathbb{F}_q$-vector spaces, and dimension is the tuple of dimensions. The footnote in [10] about "J may contain only variables on the same field" is the same issue surfacing in the algorithm;
(c) a subgroup of codimension one is the kernel of a single homomorphism to $\mathbb{F}_q$, i.e. the solution set of one linear equation;
(d) "$\mathrm{pr}_{1..k}(L)$ full and $(\bar{b}, 0) \notin L$ imply $\dim L = k$".

Item (b) is the one to be careful about: it is the reason Theorem 7.8 is stated with mixed primes $q_1, \dots, q_k$ and the reason (c) must be proved prime by prime. In Mathlib the substrate exists (`ZMod`, `Module` over `ZMod p`) but the mixed-prime dimension notion and the codimension-one characterisation are not there; see §12.

7.4 The supporting lemmas

The lemmas invoked above, in dependency order.

Lemma 7.10 ([9, Lem. 6.1]). *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*

Lemma 7.11. *Suppose $D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$, every $D^{(1)}_x$ is* S*-free, $T \in \{\text{PC}, \text{L}, \text{D}\}$, $D^{(1)} \lhd\lhd\lhd D$, and $D^{(2)}_x \leq_{\mathcal{M}T} D^{(1)}_x$ is a minimal $\mathcal{M}T$ subuniverse for every x. Then either $C^{(2)} = \varnothing$ for some constraint C, or $\mathcal{I}^{(2)}$ is 1-consistent.*

Lemma 7.12. *A constraint crucial in a reduction of a cycle-consistent irreducible instance is rectangular, and its* Con_1*'s are irreducible congruences.*

Lemma 7.13. *A relation whose restriction along strong subsets becomes empty yields a bridge between the witnessing congruences.*

Lemma 7.14 (Connected instances). *Let $\mathcal{I}$ be a cycle-consistent connected instance. Then*

(a) *any two constraints with a common variable are adjacent;*
(b) *for constraints C_1, C_2, variables $x_i \in \mathrm{Var}(C_i)$, and any path from x_1 to x_2, there is a bridge δ from $\mathrm{Con}_1(C_1, x_1)$ to $\mathrm{Con}_1(C_2, x_2)$ with $\widetilde{\delta}$ containing all pairs connected by that path;*
(p) *if $\mathcal{I}$ is linked then $\mathrm{Con}_1(C, x)$ is a perfect linear congruence for every $C \in \mathcal{I}$, $x \in \mathrm{Var}(C)$.*

Formalization note 7.15. Clause (p) is the bridge between the bridge machinery and the linear algebra; it is what makes Theorem 7.8 possible. Its proof is (b) followed by Lemma 6.22: linkedness of the instance makes $\widetilde{\delta}$ linked as a relation, which upgrades the bridge. The two senses of "linked" (Hazard 3.12) meet exactly here, and the step where one becomes the other should be isolated.

Lemma 7.16 ([11, Lem. 5.6]). *For a 1-consistent instance and a reduction, there is a tree-covering whose projection onto a given variable computes the maximal 1-consistent subreduction at that variable.*

Lemma 7.17. *Let $D^{(1)}$ be a 1-consistent reduction of a cycle-consistent instance $\mathcal{I}$ with $D^{(1)} \lhd\lhd\lhd D$, and $B <^{D_x}_{T} D^{(1)}_x$ with $T \in \{\text{BA}, \text{C}, \text{PC}\}$. Then there is a nonempty 1-consistent reduction $D^{(2)} \lhd\lhd\lhd D^{(1)}$ with $D^{(2)}_x \subseteq B$. Moreover, if $T \in \{\text{BA}, \text{C}\}$ then $D^{(2)} \leq_T D^{(1)}$; and if $T =$ PC and every $D^{(1)}_y$ is* S*-free then $D^{(2)} \leq_{\mathcal{M}\text{PC}} D^{(1)}$.*

Lemma 7.18. *A constraint that becomes empty under a minimal multi-type reduction, in an instance crucial there, has the parallelogram property; and in the linear case its first induced congruence σ satisfies $\sigma^* = D^2$.*

Lemma 7.19 (Bridges from a subdirect PC/linear instance). Suppose

(i) $\mathcal{I}$ has subdirect solution set;
(ii) $D^{(1)}$ is a reduction with $D^{(1)}_x \lhd\lhd\lhd D_x$ for every x;
(iii) $C \in \mathcal{I}$ is a constraint of type $T \in \{\text{PC}, \text{L}\}$;
(iv) $B <^{D_z}_{\mathcal{T}(\xi)} D^{(1)}_z$ for some variable z, with $\mathcal{T} \in \{\text{BA}, \text{C}, \text{S}, \text{PC}, \text{L}\}$;
(v) if $T =$ PC then $\mathcal{T} \in \{\text{PC}, \text{L}\}$;
(vi) $\mathcal{I}^{(1)}$ has a solution;
(vii) $\mathcal{I}^{(1)}$ has no solution with $z \in B$;
(viii) weakening C in $\mathcal{I}$ gives an instance with a solution in $D^{(1)}$ having $z \in B$.

Then $\mathcal{T} = T$, and for every variable x of C there is a bridge δ from ξ to $\mathrm{Con}_1(C, x)$ with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$.

Formalization note 7.20 (The longest proof in the section)**.** Lemma 7.19 carries eight numbered hypotheses and its proof runs to roughly ten pages. It is the only source of bridges in §7, and every use of connectedness routes through it. Its conclusion has two independent parts — "$\mathcal{T} = T$" and "there is a bridge with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$" — and different call sites use different parts; splitting it into two lemmas is tempting but the proof produces them together.

A formalization should attack this lemma *last* among the §3.2 material and *first* among the things budgeted generously. It, together with Theorem 6.13 (which it consumes) and Theorem 7.1 (which consumes it), is where the majority of the effort will go.

Corollary 7.21. *In the situation of Theorem 7.1(2) with $\mathcal{I}'$ satisfying (1b), the type $\mathcal{T}$ of the reduction and the type of the constraints must agree, which contradicts $\mathcal{T} \in \{\textsc{ba}, \textsc{c}\}$.*

8 The algorithm

[10] proves the correctness statements but does not contain the algorithm; that is in [9]. This section states it, because target (T1) of §1 is a proposition *about* it and cannot be written down otherwise.

8.1 Solve

```
function Solve(I):
    repeat
        I := ForceConsistency(I)
        if I = false: return "No"
        if some D_x has a strong subset B:
            I := ReduceDomain(I, x, B)
    until nothing changed
    return SolveLinear(I)
```

ForceConsistency enforces cycle-consistency and irreducibility, returning `false` if some domain empties. The reduction step is licensed by Theorem 7.6: restricting to a strong subset cannot destroy solvability. When no strong subset remains, Lemma 6.2 together with Lemma 4.14 gives, for each domain of size at least 2, a congruence σ_x with $D_x/\sigma_x \cong \mathbb{Z}_{q_1} \times \cdots \times \mathbb{Z}_{q_{n_x}}$; choose σ_x minimal. That case is SolveLinear.

8.2 SolveLinear

Write $\phi\colon \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k} \to D_{x_1}/\sigma_{x_1} \times \cdots \times D_{x_m}/\sigma_{x_m}$ for a linear map and

$$\phi^{-1}(\mathcal{I}) = \{\bar{a} : \mathcal{I} \text{ has a solution in } \phi(\bar{a})\}.$$

Computing $\phi^{-1}(\mathcal{I})$ would solve $\mathcal{I}$; the algorithm cannot do that, but it can do four things:

- **(p0)** decide $\bar{a} \in \phi^{-1}(\mathcal{I})$, by reducing each domain to the corresponding block and recursing on a smaller domain;
- **(p1)** decide $\phi^{-1}(\mathcal{I}) = \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k}$, by testing $\bar{0}$ and the k unit vectors with (p0) — correct because $\phi^{-1}(\mathcal{I})$ is a subgroup;
- **(p2)** compute $\phi^{-1}(\mathcal{I})$ when $\mathcal{I}$ is not linked, by splitting into linked pieces, recursing, and taking the union;
- **(p3)** compute $\phi^{-1}(\mathcal{I})$ when it is empty or of codimension one, by finding $\bar{a} \notin \phi^{-1}(\mathcal{I})$, then for each i a b_i with $\bar{a}[i \mapsto b_i] \in \phi^{-1}(\mathcal{I})$ if one exists, and reading off the equation $\sum_{i \in J}(y_i - a_i)/(b_i - a_i) = 1$ over the i for which b_i exists.

```
function SolveLinear(I):
    m := dim(D_{x_1}/s_1 x ... x D_{x_n}/s_n)
    phi := a bijective linear map Z_{p_1}x...xZ_{p_m} -> that product
    while phi^-1(I) != Z_{p_1}x...xZ_{p_m}:              // (p1)
        I' := I
        for C in I':                                   // drop what can be dropped
            if phi^-1(I' \ {C}) != Z_{p_1}x...xZ_{p_m}:   // (p1)
                I' := I' \ {C}
        F := phi^-1(I')                                  // (p2) or (p3)
        if F = {}: return "No"
        m := dim F ;  psi := a bijective linear map onto F ;  phi := phi o psi
    return "Yes"
```

The inner loop makes $\mathcal{I}'$ minimal with $\phi^{-1}(\mathcal{I}')$ not full. At that point Theorem 7.8 applies — its hypothesis (vii) is exactly minimality — so $\phi^{-1}(\mathcal{I}')$ is empty or of codimension one, and (p3) computes it; or $\mathcal{I}'$ is not linked and (p2) does. Since $\phi^{-1}(\mathcal{I}) \subseteq \phi^{-1}(\mathcal{I}')$, the image of ϕ shrinks while still containing all solutions. The dimension strictly decreases, so the loop terminates.

8.3 What is honestly formalizable

Formalization note 8.1 (Three obstacles between the pseudocode and a Lean function). (a) *Termination is not structural.* SOLVE recurses through (p0) on strictly smaller domains, and SOLVELINEAR loops on a strictly decreasing dimension. Neither is a structural recursion; both need an explicit well-founded measure, and the two interleave — SOLVELINEAR calls SOLVE on a smaller domain, which calls SOLVELINEAR again. The measure is lexicographic: $(\sum_x |D_x|,\ \dim\phi)$.

(b) *Several steps quantify over objects that are only finitely many by accident.* "If some D_x has a strong subset B" quantifies over subsets of D_x, over congruences on D_x, and — through absorption — over *term operations*, of which there are infinitely many. Decidability comes from the fact that on a finite set there are finitely many operations of each arity and finitely many subsets, so "B absorbs A with some term" is decidable by exhaustive search over Clo_k for $k \leq |A|^{|A|}$; but that bound is a theorem, not a definition, and it is the only reason the algorithm is an algorithm.

(c) *Polynomial time is a different subject.* The bound in [9] counts calls and bounds the recursion depth by $|A|$; making that formal needs a cost model. See §11.

Theorem 8.2 (Target (T1)). *There is a function SOLVE from instances of* CSP(Γ) *to Booleans such that for every instance $\mathcal{I}$,*

$$\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

Theorem 8.2 mentions no machine, is provable from §7, and is the honest formal content of "the algorithm works". It is strictly stronger than decidability of CSP(Γ), which is trivial for a finite template: it says that *this* algorithm — the one whose existence is the dichotomy — is correct.

9 The hard half

If no weak near-unanimity operation preserves Γ, then $\mathrm{CSP}(\Gamma)$ is NP-complete. This half is older, shorter, and better understood than the tractable half, and it factors cleanly into algebra and complexity.

9.1 The chain

(1) *Reduction to a core, and to an idempotent algebra.* One may assume Γ contains all singleton unary relations, so that $\mathrm{Pol}(\Gamma)$ is idempotent, at the cost of a polynomial-time reduction.

(2) *Maróti–McKenzie.* A finite idempotent algebra has a Taylor term if and only if it has a weak near-unanimity term [6].

(3) *No Taylor term ⇒ pp-interpretation.* If a finite idempotent algebra has no Taylor term, then the variety it generates contains a two-element algebra whose only operations are projections; equivalently, Γ primitively positively interprets every finite constraint language, in particular a template for 3-Sat or Nae-3-Sat.

(4) *Bulatov–Jeavons–Krokhin.* A primitive positive interpretation of Δ in Γ yields a polynomial-time many-one reduction $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$ [1].

(5) *Cook–Levin.* Nae-3-Sat is NP-hard.

Steps (1)–(3) are algebra; step (4) is a syntactic construction plus a complexity-theoretic wrapper; step (5) is pure complexity theory.

9.2 The layered targets

Theorem 9.1 (Target (H0))**.** *Let* $\mathbf{A}$ *be a finite idempotent algebra with no weak near-unanimity term operation. Then* $\mathrm{Inv}(\mathbf{A})$ *primitively positively interprets the constraint language* $\{\textsc{Nae}\}$ *on* $\{0, 1\}$*, where* $\textsc{Nae} = \{0, 1\}^3 \setminus \{(0, 0, 0), (1, 1, 1)\}$*.*

Theorem 9.2 (Target (H1))**.** *If* Γ *primitively positively interprets* Δ *then* $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$*. Consequently, under* (H0)*,* $\mathrm{CSP}(\Gamma)$ *is* NP*-hard.*

Theorem 9.1 mentions no machine. Theorem 9.2 is where the complexity layer enters, and it is small — the reduction is a syntactic substitution of formulas — but it cannot be stated at all without a formalized notion of polynomial-time many-one reduction.

9.3 Definitions

Definition 9.3 (pp-formula, pp-definability)**.** A *primitive positive formula* over Γ is $\exists y_1 \ldots \exists y_s \bigwedge_j R_j(\bar{z}_j)$ with $R_j \in \Gamma \cup \{=\}$. A relation is *pp-definable* from Γ if some pp-formula defines it.

Definition 9.4 (pp-interpretation)**.** Γ on A *pp-interprets* Δ on B if there are $d \geq 1$, a pp-definable $F \subseteq A^d$, and a surjection $\pi \colon F \to B$ such that the π-preimage of every relation of Δ, and of the equality relation on B, is pp-definable from Γ.

Theorem 9.5 (Geiger; Bodnarchuk–Kalužnin–Kotov–Romov)**.** *For a finite A, a relation is pp-definable from* Γ *if and only if it is preserved by every polymorphism of* Γ*.*

Formalization note 9.6 (The Galois connection is the expensive import)**.** Theorem 9.5 is what converts the algebraic hypothesis ("no WNU polymorphism") into a syntactic conclusion ("pp-defines a hard relation"), and it is the only genuinely substantial import of §9. The easy direction is a one-line calculation. The hard direction — every relation preserved by all polymorphisms is pp-definable — goes through the free algebra on $|A|^k$ generators and an indicator-instance construction; it is roughly four printed pages and is the single largest item in the (H0) budget. It is also of independent value and would be a reasonable Mathlib contribution on its own.

Remark 9.7 (Why the hard half is not the hard half)**.** Despite its name, (H0) is far smaller than (T0): perhaps twenty printed pages against a hundred. The asymmetry is intrinsic. Showing a problem is hard requires exhibiting one construction; showing it is easy requires an algorithm that works on every instance, and the correctness proof must survive every configuration the instance can present.

10 Defects in the sources

A blueprint earns its keep by finding the places where the source cannot be transcribed. This section collects them. They are not criticisms of [10], which is unusually careful prose — across nine thousand lines of TEX it says "obvious" three times and "clearly" never. They are the residue that any informal exposition leaves, and each one is a decision a formalization has to make before it can write the corresponding statement.

Items marked **blocking** must be resolved before the affected statement can be written in Lean at all. Items marked **convention** are places where the source is consistent but leaves the reader to supply a reading.

10.1 Blocking

D1 Lemma 6.10 is false as printed. [10, Lemma 19] drops two hypotheses present in its cited source: subdirectness of R in the first coordinate, and a common absorbing term in the BA case. Counterexample and repair in Hazard 6.11. Used three times in §3 and three times in §4 of the source.

D2 Corollary 6.6 is never proved. It is stated in §2.3, used four times inside §5 and once in §3, and neither proved nor restated anywhere. It is Lemma 6.4 specialised to the canonical surjection $\mathbf{A} \to \mathbf{A}/\delta$, but the source does not say so, and the specialisation is not quite mechanical: clause (m)'s hypothesis "B/δ is S-free" has to be matched against clause (fm)'s "$f(B)$ is S-free".

D3 There is a citation cycle in §5. Tracing the proofs: Lemma 9 ⇒ Lemma 7 ⇒ Lemma 86 ⇒ Lemma 85 ⇒ Corollary 22 ⇒ Theorem 21 ⇒ Lemma 8 ⇒ Lemma 9, closed where Lemma 85's $T = \mathrm{C}$ case invokes Corollary 22. Nothing in §5.4 or §5.7 can be formalized until the cycle is broken. The break appears to require a mixed essentiality lemma for three distinct central subuniverses, which is not in the paper.

D4 Theorem 6.13(c) asserts $m = 2$ without argument. The proof reduces to pairs and shows the types agree; for PC, $\sigma_1 = \sigma_2$ forces $m = 2$. Nothing rules out $m \geq 3$ with all types C. A Lean statement copying the source is unprovable as it stands. Same fix as D3.

D5 Theorem 7.8 hypothesis (ii) is too weak for its proof. The source says "$\mathbf{D}_{x_i}$ is S-free"; the proof, and the informal claim it formalises, need "no nontrivial binary absorbing or central subuniverse on $\mathbf{D}_{x_i}$" — which is the *commented-out* gloss sitting immediately below the hypothesis in the source. S-freeness is strictly weaker (Hazard 4.7). Strengthen the hypothesis.

D6 Lemma 7.14(a) applies Lemma 7.13 without one of its hypotheses. The missing hypothesis can fail; a witness is $R = \{(x, z, y) \in \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_4 : x \equiv z \equiv y \pmod 2\}$. In [9] the needed tuples came from *criticality*, a notion [10] dropped. The fix is to add "every constraint relation is critical" to the definition of a connected instance — available, since crucial implies critical — at the cost of reimporting about a page of the older machinery.

D7 Two gaps in Theorem 7.1. In Case 1 the inductive hypothesis yields (1c) relative to $D^{(2)}$ while the goal is (1c) relative to $D^{(1)}$; the source writes only "we derive the required conditions". And both endgames conclude "(1b) if the solution set is subdirect, (1c) otherwise", but (1c) demands a *linked* connected subinstance where only connectedness was established — linkedness has to be extracted from irreducibility, which needs a fragmentation descent and a separate treatment of the empty-solution-set case.

D8 A lemma the proof needs is commented out of the source. `LEMMinimalContainingIsMinimal` — that an inclusion-minimal $\mathcal{M}T$-subset containing a given element is minimal among $\mathcal{M}T$-subsets — is commented out along with its citation, yet is genuinely used in Case 2 of Theorem 7.1. It is true and must be reproved.

D9 **The algorithm's pseudocode does not match the theorem it invokes.** SOLVE-LINEAR *removes* constraints, whereas Theorem 7.8(vii) is about *weakening* them; and primitive (p2) returns a union of affine subspaces, which the next line treats as affine. [10] labels its pseudocode a sketch and points to [9] for the precise algorithm, so this is a defect of the sketch rather than of the proof — but target (T1) is a statement about the algorithm, so it must be repaired from [9] before (T1) can be stated.

D10 **The arity in Lemma 6.20 is suspect.** The imported statement gives a special WNU of arity $n^{n!}$; the period of $y \mapsto w(x, \ldots, x, y)$ need not divide $n^{n!}$ — take $n = 3$ and an element of order 5. The remedy is to weaken the statement to "*some* arity N", which is all the dichotomy proof uses. The precise arity matters only for §4 of the source, which is cut.

10.2 Convention

C1 $\mathbf{Z}_p \in \mathcal{V}_n$ requires $p \mid n-1$ (Hazard 3.3).

C2 σ^* is a tolerance, not a congruence (Hazard 3.22). Typing it as a congruence collapses the linear/PC distinction silently.

C3 The empty set is a subuniverse of every idempotent algebra, and is vacuously absorbing and vacuously central. Read syntactically, the clause defining $<_{\mathrm{s}}$ is therefore universally true. The global convention "we do not allow empty subuniverses" repairs it, but the clause does not inherit that convention on its own (Convention 2.4).

C4 Whether $\sigma = A^2$ counts as irreducible is genuinely ambiguous — it turns on whether the empty intersection is allowed — and σ^* does not exist in that case. [10] drops the word "proper" that [9] has. Legislating that the empty family *is* allowed makes irreducibility imply properness, which is the convenient choice and is what the Lean development adopts.

C5 $\lhd\lhd\lhd$ is data, not merely a proposition: several §5 proofs speak of "the congruences coming from $C \lhd\lhd\lhd A$" and of minimal chain length (Hazard 4.21).

C6 "Dimension" of $\prod \mathbb{Z}_{q_i}$ with distinct primes is never defined, and there is no field over which it is a vector space. Adopt composition length and prove additivity (Hazard 7.9).

C7 Two inequivalent definitions of "linked instance" coexist: global connectivity of the value graph, in Informal Claim (IC3); and the per-variable condition in §3.1. The second does not imply the first. Use the formal one and carry non-fragmentation separately (Hazard 3.12).

C8 B/σ is the image of B in A/σ, not the quotient of B; and $(C_1 \cap \cdots \cap C_t)/\delta = \bigcap C_i/\delta$ is false in general and is proved on the fly inside Lemma 6.4(fm) (Hazard 3.14).

C9 Three sentences beginning "Notice that" are used as lemmas and never proved: that the relation S of Definition 4.8(c) is a bridge with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$; that σ is irreducible/linear/PC exactly when $0_{\mathbf{A}/\sigma}$ is; and that the two formulations of S-freeness agree. Add Lemma 3.16 to the same list.

C10 One citation silently drops a case: the source's use of [11, Thm. 6.15] omits a third alternative — a nontrivial *projective* subuniverse — present in the original. Its elimination is legitimate under the standing Taylor hypothesis, but the source does not say so, and a formalization must state it as a hidden import and check it against what the predecessor development actually proves.

Remark 10.1 (What this list is evidence of)**.** Ten blocking items in a fifty-page paper is not a high rate; it is roughly what one should expect, and it is the reason blueprints exist. Note the distribution: one is a genuine misstatement (D1), one is an omission (D2), two are gaps in the mathematics that need new arguments (D3, D4), three are hypotheses that are too weak or missing (D5, D6, D8), and three are artefacts of the interface with the older paper (D7, D9, D10). None of them threatens the theorem. All of them cost time.

11 The complexity layer

Targets (T2) and (H1) are stated here and not proved. The point of stating them is to be precise about what is *not* being claimed.

Definition 11.1 (What a cost model would have to supply)**.** To state (T2) one needs: an encoding of instances of CSP(Γ) as strings; a machine model with a step count; a predicate "f is computed in time bounded by a polynomial in the input length"; and closure of that predicate under composition. To state (H1) one needs, in addition, a polynomial-time many-one reduction relation $\leq_p$ and the class NP.

Mathlib supplies the machine (`Turing.TM2`), an encoding (`Computability.Encoding`), and a time-bounded computation predicate (`Turing.TM2ComputableInPolyTime`). It does not supply a complexity *class*, P, NP, NP-completeness, $\leq_p$, or Sat; and the composition lemma `TM2ComputableInPolyTime.comp` is an open `proof_wanted` in Mathlib itself. Moreover `TM2ComputableInPolyTime` applies to total functions $\alpha \to \beta$, never to languages, so even the existing predicate would need reshaping.

Remark 11.2. Building this layer is a substantial project — comparable in size to the algebraic core, and with a large formal-verification literature of its own — and it contributes none of the mathematics of the dichotomy. Attempting it as part of this project would be a scoping error. It is exactly the kind of component that should be built once, in Mathlib, by people interested in complexity theory, and then consumed.

12 Formalization

12.1 What is reused

The predecessor development `ZhukLean` is a complete, `sorry`-free Lean 4 formalization of Zhuk's centre theorem, built on `Mathlib.ModelTheory`. It is consumed here as a `lake` dependency, not copied. It supplies:

- products of structures (`piStructure`, `prodStructure`, coordinatewise realization, reindexing and evaluation homomorphisms) — the one part of the universal-algebra vocabulary Mathlib lacks;
- absorption in the tuple form (`Witnesses`, `Absorbs`, `BinAbsorbs`), Taylor terms (`IsTaylorOn`), and central absorption (`CentrallyAbsorbs`);
- the relational description of absorption by essential relations (Barto–Kazda), which is what converts absorption at some arity into absorption at arity 3;
- Zhuk's centre theorem itself (`zhuk_center`) and the ternary collapse (`exists_ternary_witnesses`) — which is precisely Lemma 4.4, cited without proof in [10].

Three of [10]'s sixteen black-box imports are therefore already discharged.

12.2 What is built here

Module	Contents	Status
`CSP/Product`	products of structures	complete
`CSP/Congruence`	`Congruence L M`, order, `sInf`, quotient	complete
`CSP/Irreducible`	irreducibility, existence and uniqueness of σ^*	complete
`CSP/Bridge`	bridges, collapse, composition, linear/PC	2 imports open
`CSP/Types`	the six types, multi-types, $\lhd\lhd\lhd$	complete
`CSP/Instance`	instances, reductions, consistency, cruciality	complete
`CSP/Main`	the main statements	targets

Everything below `CSP/Main` is `sorry`-free. The two open items in `CSP/Bridge` are Lemma 3.26 and its collapse identity, both imported from [9].

12.3 Design decisions

Build on `Mathlib.ModelTheory`, keep the language generic. A bespoke "finite algebra with one WNU operation" type would bundle the standing hypotheses as fields and avoid threading them, which is a real cost in the predecessor development. But it would discard `Substructures.lean` — 985 lines of exactly the Sg API needed, including `mem_closure_iff_exists_term` with the variable type equal to the generating set, an interface no hand-rolled clone matches — and it would discard the predecessor's 1600 lines wholesale. Congruences and quotients are also *cheaper* in `ModelTheory`, not dearer. Instantiating to $\mathcal{V}_n$ is a thin layer on top.

$\lhd\lhd\lhd$ is `Relation.ReflTransGen`. Its induction principle is the one the proofs use.

Dotted relations are defined as disjunctions. `DotLll C B := C = ∅ ∨ Lll C B`, so the case split is forced at every use site rather than resolved by the reader.

The bridge criterion is the definition of linearity. Zhuk defines a linear congruence by a per-block isomorphism to $\mathbb{Z}_p^m$ and derives the bridge criterion. We reverse this: the criterion is first-order over finite data and is what every consumer uses; the block description becomes a structure theorem. See Hazard 4.10.

σ^* is data attached to irreducibility. `Irreducible.exists_isCover` produces it; there is no standalone function, because there is no cover without irreducibility. A pleasant by-product: with the definition taken literally, irreducibility *implies* $\sigma \neq A^2$, since the empty family intersects to the universe. The source never says this.

Instances are lists; scopes are tuples; the variable type is a parameter. See Hazards 5.2 and 5.16. The last of these is the decision that has not yet been cashed out, and it should be made before anything about coverings is written.

Nothing is stubbed with `sorry` at the level of a definition. `IsConnectedInstance` and `IsExpandedCovering` are *absent* rather than defined as `sorry`, because a `def : Prop := sorry` is a junk constant that downstream proofs can silently lean on. Absence forces the design decision; a stub hides it.

12.4 Order of attack

1. Lemma 6.10. Small, self-contained, used everywhere, and adjacent to what is already formalized.
2. The mixed-prime linear algebra of Hazard 7.9. Independent of everything else; can be done in parallel.
3. Lemma 3.26. Twelve source lines of relational algebra; closes the two open items in `CSP/Bridge`.
4. The coproduct design for coverings, then Proposition 5.15.
5. Theorem 6.13. The one hard statement in the interface, and the sole source of bridges.
6. Lemma 7.19. Eight hypotheses, ten pages.
7. Theorem 7.1. Last.

12.5 Scope

The calibration point is that the predecessor formalized about 3.5 printed pages in about 1670 lines of Lean — roughly 480 lines per page. Excluding §11 and [10]'s §4, which is an independent second result and not needed for the dichotomy, the transitive closure of what must be proved is on the order of 100–130 printed pages, so 35 000–60 000 lines of Lean.

That is a multi-person-year project, and it should be described as one. What is finished here is its design: the route, the statements, the vocabulary at formalization granularity, the hazard list, the missing-layer inventory, and a building development whose foundations are proved and whose summit is stated.

References

[1] A. Bulatov, P. Jeavons, A. Krokhin, *Classifying the complexity of constraints using finite algebras*, SIAM J. Comput. **34** (2005), 720–742.

[2] Z. Brady, *Notes on CSPs and polymorphisms*, arXiv:2210.07383.

[3] A. Bulatov, *A dichotomy theorem for nonuniform CSPs*, arXiv:1703.03021; FOCS 2017.

[4] T. Feder, M. Y. Vardi, *The computational structure of monotone monadic SNP and constraint satisfaction*, SIAM J. Comput. **28** (1999), 57–104.

[5] M. Istinger, H. K. Kaiser, *A characterization of polynomially complete algebras*, J. Algebra **56** (1979), 103–110.

[6] M. Maróti, R. McKenzie, *Existence theorems for weakly symmetric operations*, Algebra Universalis **59** (2008), 463–489.

[7] L. Barto, Z. Brady, A. Bulatov, M. Kozik, D. Zhuk, *Minimal Taylor algebras as a common framework for the three algebraic approaches to the CSP*, LICS 2021.

[8] R. Willard, slides on bridges, similarity and centralizers.

[9] D. Zhuk, *A proof of the CSP dichotomy conjecture*, J. ACM **67** (2020), no. 5, 1–78; arXiv:1704.01914.

[10] D. Zhuk, *A simplified proof of the CSP dichotomy conjecture and XY-symmetric operations*, arXiv:2404.01080v2.

[11] D. Zhuk, *Strong subalgebras and the constraint satisfaction problem*, J. Mult.-Valued Logic Soft Comput. **36** (2021); arXiv:2005.00593.